CHAPTER 18

Handling Overlapping Points

When we want to visualize large or very large datasets, we often experience the challenge that simple *x*–*y* scatterplots do not work very well because many points lie on top of each other and partially or fully overlap. And similar problems can arise even in small datasets if data values were recorded with low precision or rounded, such that multiple observations have exactly the same numeric values. The technical term commonly used to describe this situation is *overplotting*, which means that we are plotting many points on top of each other. Here I describe several strategies you can pursue when encountering this challenge.

Partial Transparency and Jittering

We first consider a scenario with only a moderate number of data points but with extensive rounding. Our dataset contains fuel economy during city driving and engine displacement for 234 popular car models released between 1999 and 2008 (Figure 18-1). In this dataset, fuel economy is measured in miles per gallon (mpg) and is rounded to the nearest integer value. Engine displacement is measured in liters and is rounded to the nearest deciliter. Due to this rounding, many car models have exactly identical values. For example, there are 21 cars total with 2.0 liter engine displacement, and as a group they have only four different fuel economy values: 19, 20, 21, or 22 mpg. Therefore, in Figure 18-1 these 21 cars are represented by only four distinct points, so that 2.0 liter engines appear much less popular than they actually are. Moreover, the dataset contains two four-wheel drive cars with 2.0 liter engines, which are represented by black dots. However, these black dots are fully occluded by yellow dots, so that it looks like there are no four-wheel drive cars with a 2.0 liter engine.

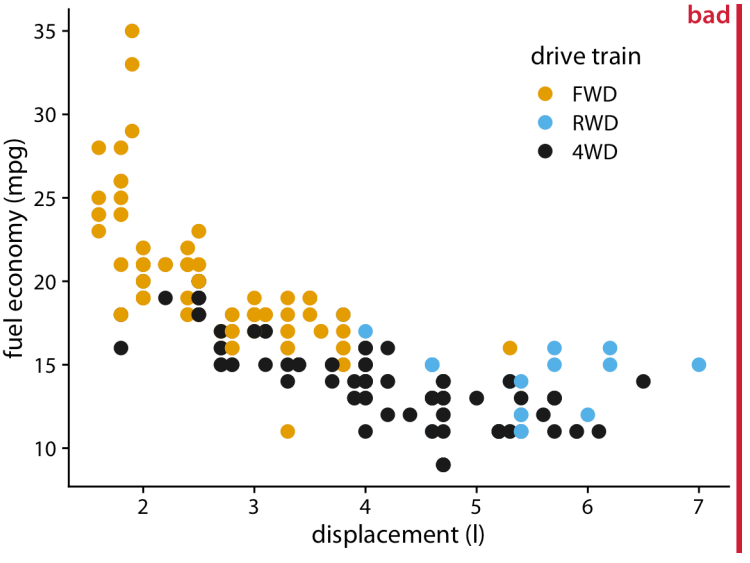


Figure 18-1. City fuel economy versus engine displacement, for popular cars released between 1999 and 2008. Each point represents one car. The point color encodes the drive train: front-wheel drive (FWD), rear-wheel drive (RWD), or four-wheel drive (4WD). The figure is labeled "bad" because many points are plotted on top of others and obscure them. Data source: US Environmental Protection Agency (EPA), https://fueleconomy.gov.

One way to ameliorate this problem is to use partial transparency. If we make individual points partially transparent, then overplotted points appear as darker points and thus the shade of the points reflects the density of points in that location of the graph (Figure 18-2).

However, making points partially transparent is not always sufficient to solve the issue of overplotting. For example, even though we can see in Figure 18-2 that some points have a darker shade than others, it is difficult to estimate how many points were plotted on top of each other in each location. In addition, while the differences in shading are clearly visible, they are not self-explanatory. A reader who sees this figure for the first time will likely wonder why some points are darker than others and will not realize that those points are in fact multiple points stacked on top of each other. A simple trick that helps in this situation is to apply a small amount of jitter to the points—i.e., to displace each point randomly by a small amount in either the x or the y direction or both. With jitter, it is immediately apparent that the darker areas arise from points that are plotted on top of each other (Figure 18-3). Also, now, for the first time the black dots that represent four-wheel drive cars with 2.0 liter engines can be seen.

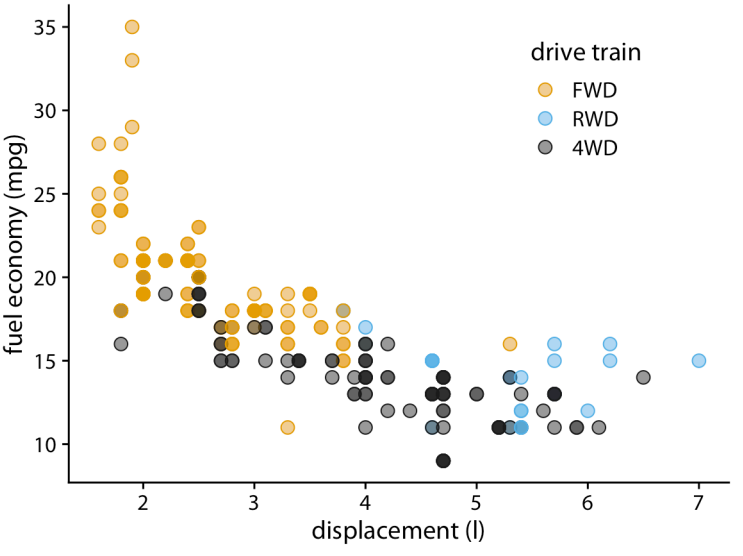


Figure 18-2. City fuel economy versus engine displacement. Because points have been made partially transparent, points that lie on top of other points can now be identified by their darker shade. Data source: EPA.

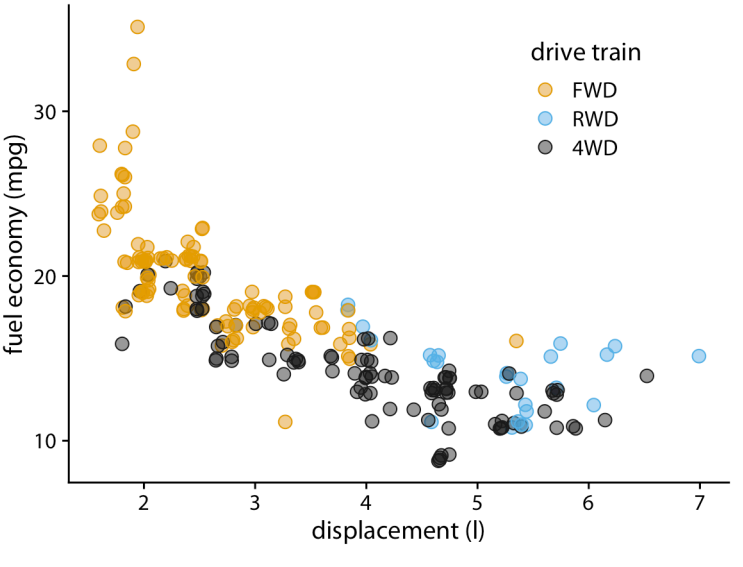


Figure 18-3. City fuel economy versus engine displacement. By adding a small amount of jitter to each point, we can increase the visibility of the overplotted points without substantially distorting the message of the plot. Data source: EPA.

One downside of jittering is that it does change the data and therefore has to be performed with care. If we jitter too much, we end up placing points in locations that are not representative of the underlying dataset. The result is a misleading visualization of the data. See Figure 18-4 as an example.

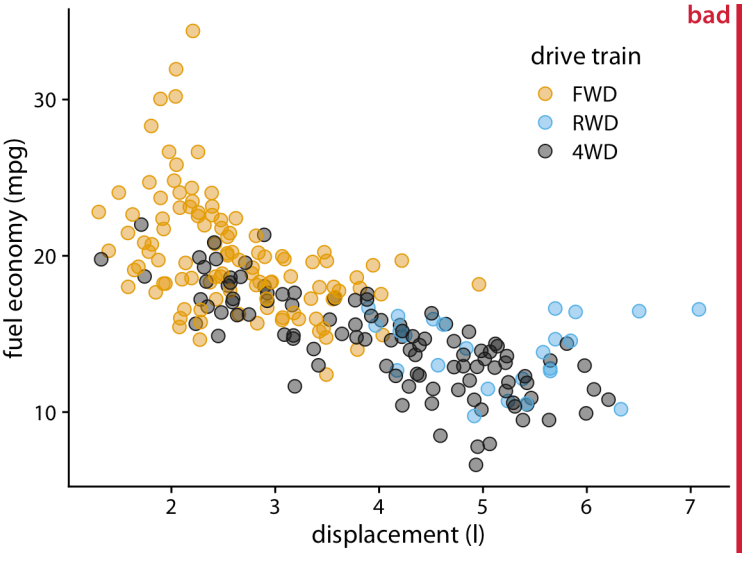


Figure 18-4. City fuel economy versus engine displacement. By adding too much jitter to the points, we have created a visualization that does not accurately reflect the underlying dataset. Data source: EPA.

2D Histograms

When the number of individual points gets very large, partial transparency (with or without jittering) will not be sufficient to resolve the overplotting issue. What will typically happen is that areas with high point density will appear as uniform blobs of dark color, while in areas with low point density the individual points are barely visible (Figure 18-5). And changing the transparency level of individual points will either ameliorate one or the other of these problems while worsening the other; no transparency setting can address both at the same time.

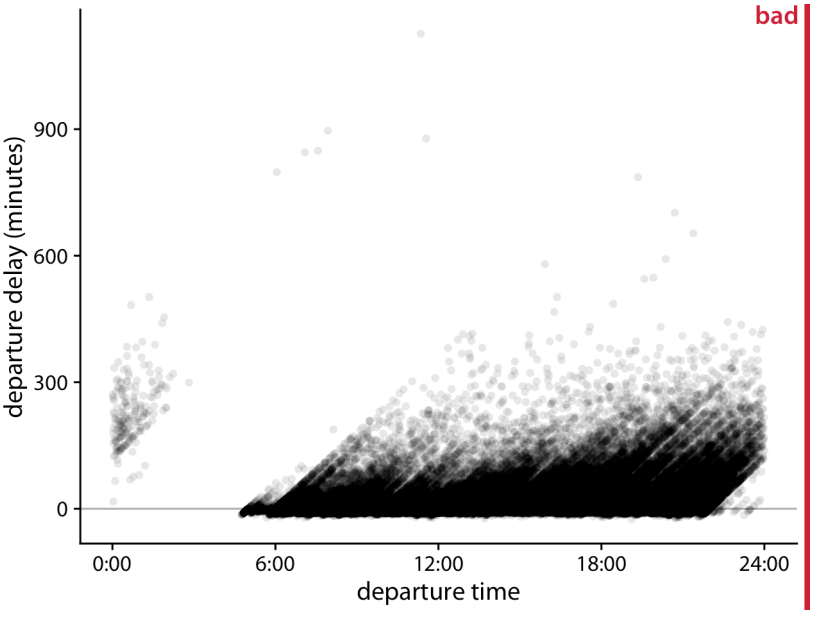


Figure 18-5. Departure delay in minutes versus flight departure time, for all flights departing Newark Airport (EWR) in 2013. Each dot represents one departure. Data source: US Dept. of Transportation, Bureau of Transportation Statistics.

Figure 18-5 shows departure delays for over 100,000 individual flights, with each dot representing one flight departure. Even though we have made the individual dots fairly transparent, the majority of them just form a black band at between 0 and 300 minutes departure delay. This band obscures whether most flights depart approximately on time or with a substantial delay (say, 50 minutes or more). At the same time, the most-delayed flights (with delays of 400 minutes or more) are barely visible due to the transparency of the dots.

In such cases, instead of plotting individual points, we can make a *2D histogram*. A 2D histogram is conceptually similar to a 1D histogram, as discussed in Chapter 7, but now we bin the data in two dimensions. We subdivide the entire x–y plane into small rectangles, count how many observations fall into each one, and then color the rectangles by those counts. Figure 18-6 shows the result of this approach for the departure delay data. This visualization highlights several important features of the flight departure data. First, the vast majority of departures during the day (from 6 a.m. to about 9 p.m.) actually depart without delay or even early (negative delay). However, a modest number of departures have a substantial delay. Moreover, the later a plane departs in the day, the more of a delay it can have. Importantly, the departure time is the actual time of departure, not the scheduled time of departure, so this figure does not necessarily tell us that planes scheduled to depart early never experience

delay. What it does tell us, though, is that if a plane departs early it either has little delay or, in very rare cases, a delay of around 900 minutes.

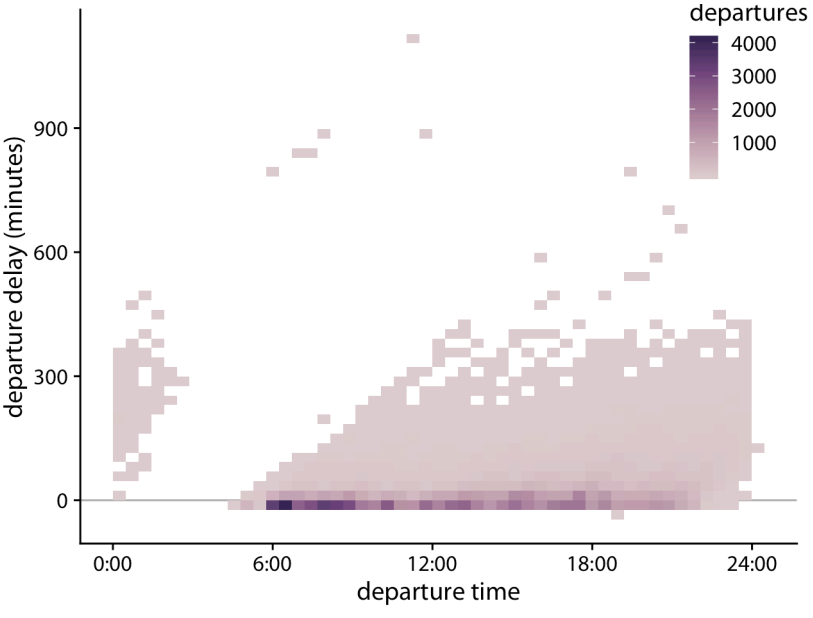


Figure 18-6. Departure delay in minutes versus the flight departure time. Each colored rectangle represents all flights departing at that time with that departure delay. Coloring represents the number of flights represented by that rectangle. Data source: US Dept. of Transportation, Bureau of Transportation Statistics.

As an alternative to binning the data into rectangles, we can bin into hexagons [Carr et al. 1987]. This approach has the advantage that the points in a hexagon are, on average, closer to the hexagon's center than the points in an equal-area square are to the center of the square. Therefore, the colored hexagons represent the data slightly more accurately than the colored rectangles. Figure 18-7 shows the flight departure data with hexagon binning rather than rectangular binning.

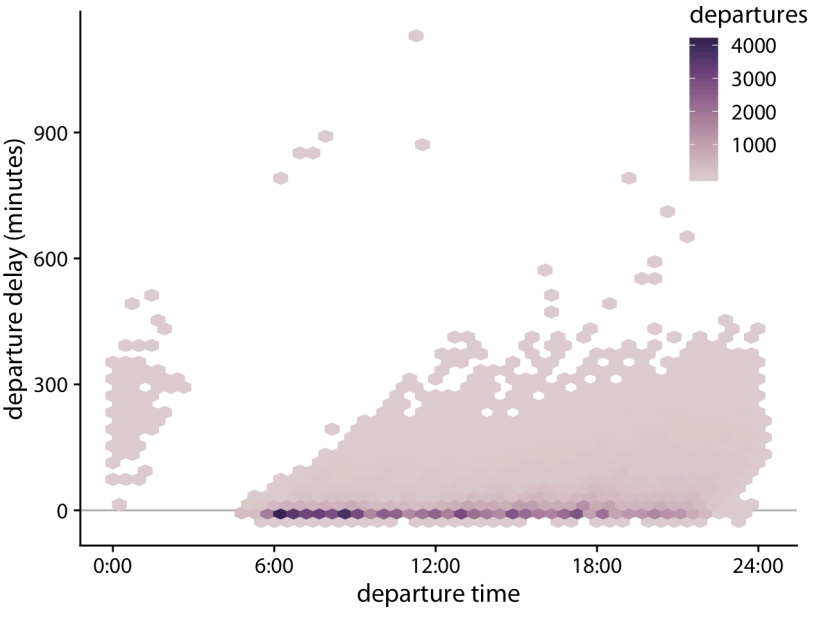


Figure 18-7. Departure delay in minutes versus the flight departure time. Each colored hexagon represents all flights departing at that time with that departure delay. Coloring represents the number of flights represented by that hexagon. Data source: US Dept. of Transportation, Bureau of Transportation Statistics.

Contour Lines

Instead of binning data points into rectangles or hexagons, we can also estimate the point density across the plot area and indicate regions of different point densities with contour lines. This technique works well when the point density changes slowly across both the *x* and the *y* dimensions.

As an example for this approach, we return to the blue jays dataset from Chapter 12. Figure 12-1 showed the relationship between head length and body mass for 123 blue jays, and there was some amount of overlap among the points. We can highlight the distribution of points more clearly by making the points smaller and partially transparent and plotting them on top of contour lines that delineate regions of similar point density (Figure 18-8). We can further enhance the perception of changes in the point density by shading the regions enclosed by the contour lines, using darker colors for regions representing higher point densities (Figure 18-9).



Figure 18-8. Head length versus body mass for 123 blue jays, as in Figure 12-1. Each dot corresponds to one bird, and the lines indicate regions of similar point density. The point density increases toward the center of the plot, near a body mass of 75 g and a head length between 55 mm and 57.5 mm. Data source: Keith Tarvin, Oberlin College.

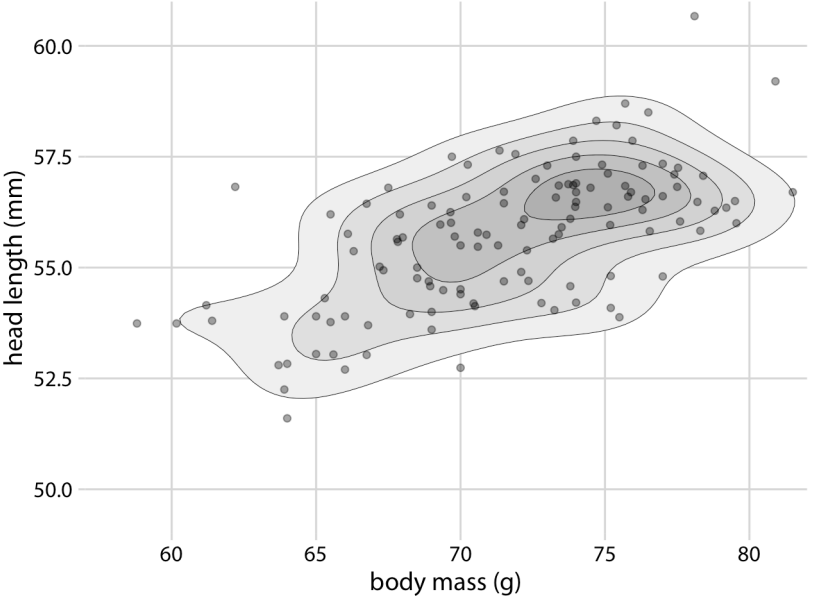


Figure 18-9. Head length versus body mass for 123 blue jays. This figure is nearly identical to Figure 18-8, but now the areas enclosed by the contour lines are shaded with increasingly darker shades of gray. This shading creates a stronger visual impression of increasing point density toward the center of the point cloud. Data source: Keith Tarvin, Oberlin College.

In Chapter 12, we also looked at the relationship between head length and body mass separately for male and female birds (Figure 12-2). We can do the same with contour lines, by drawing separately colored contour lines for male and female birds (Figure 18-10).

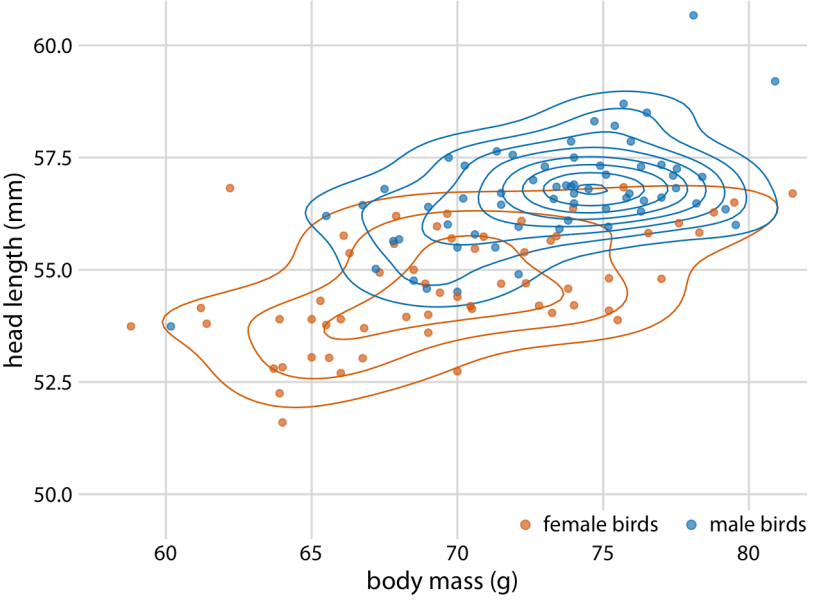


Figure 18-10. Head length versus body mass for 123 blue jays. As in Figure 12-2, we can also indicate the birds' sex by color when drawing contour lines. This figure highlights how the point distribution is different for male and female birds. In particular, male birds are more densely clustered in one region of the plot area whereas female birds are more spread out. Data source: Keith Tarvin, Oberlin College.

Drawing multiple sets of contour lines in different colors can be a powerful strategy for showing the distributions of several point clouds at once. However, this technique needs to be employed with care. It only works when the number of groups with distinct colors is small (two to three) and the groups are clearly separated. Otherwise, we may end up with a hairball of differently colored lines all crisscrossing each other and not showing any particular pattern at all.

To illustrate this potential problem, I will employ the diamonds dataset, which contains information for 53,940 diamonds, including their price, weight (carat), and cut. Figure 18-11 shows this dataset as a scatterplot. The figure exhibits severe overplotting. There are so many different-colored points on top of one another that it is impossible to discern anything beyond the overall broad outline of where diamonds fall on the price–carat spectrum.

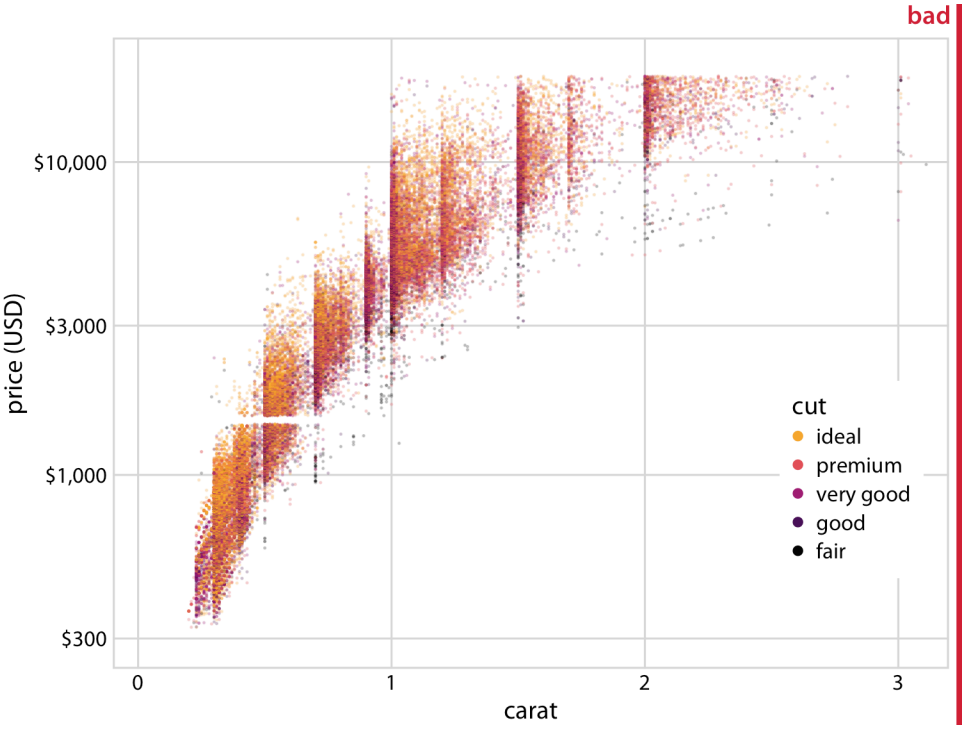


Figure 18-11. Price of diamonds versus their carat value, for 53,940 individual diamonds. Each diamond's cut is indicated by color. The plot is labeled as "bad" because the extensive overplotting makes it impossible to discern any patterns among the different diamond cuts. Data source: Hadley Wickham, ggplot2.

We could try to draw colored contour lines for the different qualities of cut, as in Figure 18-10. However, in the diamonds dataset, we have five distinct colors and the groups strongly overlap. Therefore, the contour plot (Figure 18-12) is not much better than the original scatterplot (Figure 18-11).

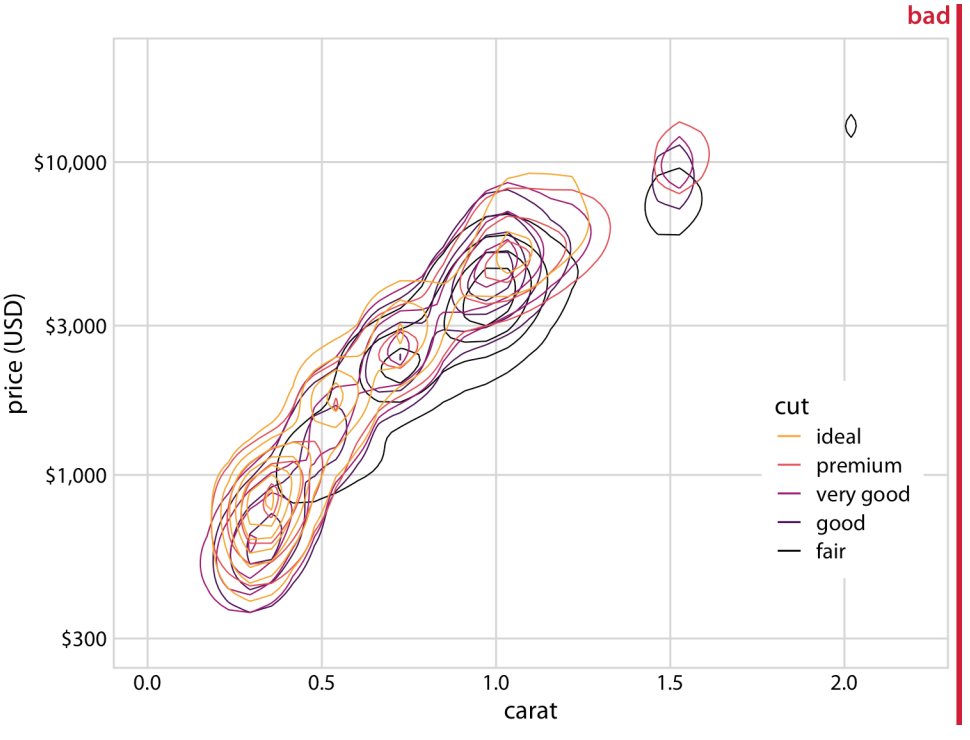


Figure 18-12. Price of diamonds versus their carat value. As in Figure 18-11, but now individual points have been replaced by contour lines. The resulting plot is still labeled "bad," because the contour lines all lie on top of each other. Neither the point distribution for individual cuts nor the overall point distribution can be discerned. Data source: Hadley Wickham, ggplot2.

What helps here is to draw the contour lines for each cut quality in its own plot panel (Figure 18-13). The purpose of drawing them all in one panel might be to enable visual comparison between the groups, but Figure 18-12 is so busy that a comparison isn't possible. Instead, in Figure 18-13, the background grid enables us to make comparisons across cut qualities by paying attention to where exactly the contour lines fall relative to the grid lines. (A similar effect could have been achieved by plotting partially transparent individual points instead of contour lines in each panel.)

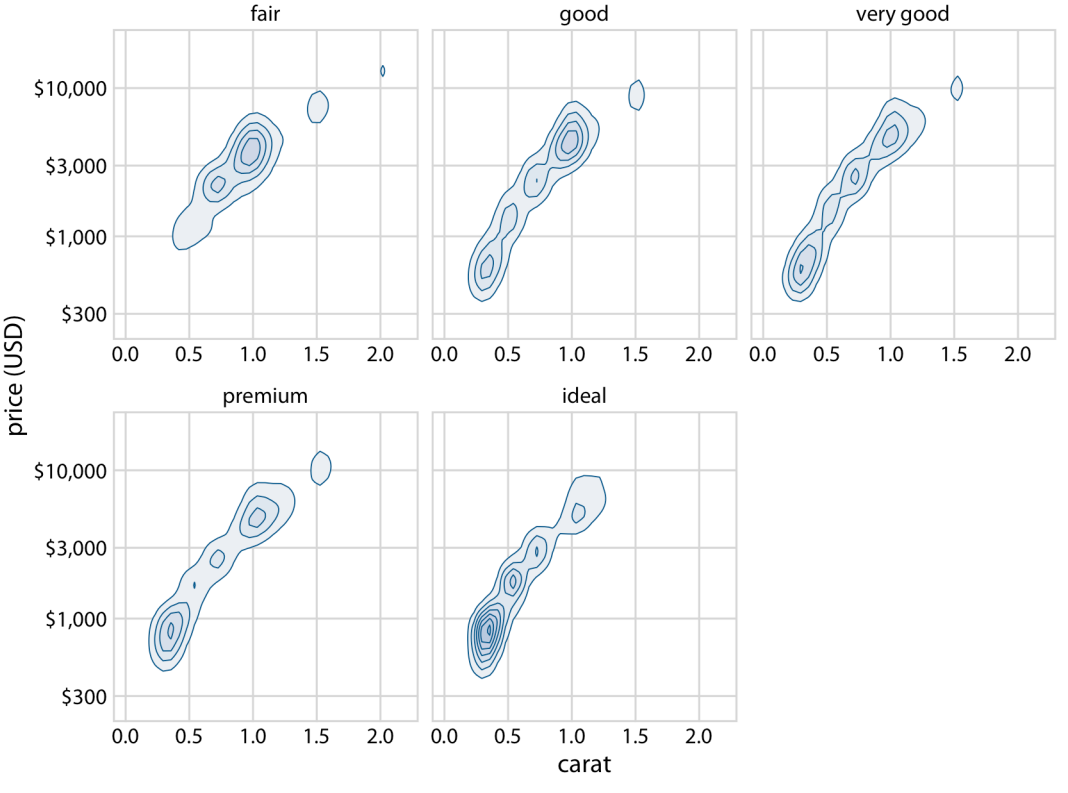


Figure 18-13. Price of diamonds versus their carat value. Here, we have taken the density contours from Figure 18-12 and drawn them separately for each cut. We can now see that better cuts (very good, premium, ideal) tend to have lower carat values than the poorer cuts (fair, good) but command a higher price per carat. Data source: Hadley Wickham, ggplot2.

We can now make out two main trends. First, the better cuts (very good, premium, ideal) tend to have lower carat values than the poorer cuts (fair, good). Recall that carat is a measure of diamond weight (1 carat = 0.2 grams). Better cuts tend to result (on average) in lighter diamonds because more material needs to be removed to create them. Second, at the same carat value, better cuts tend to command higher prices. To see this pattern, look for example at the price distribution for 0.5 carats. The distribution is shifted upwards for better cuts, and in particular it is substantially higher for diamonds with ideal cut than for diamonds with fair or good cut.

CHAPTER 19

Common Pitfalls of Color Use

Color can be an incredibly effective tool to enhance data visualizations. At the same time, poor color choices can ruin an otherwise excellent visualization. Color needs to be applied to serve a purpose, it must be clear, and it must not distract.

Encoding Too Much or Irrelevant Information

One common mistake is trying to give color a job that is too big for it to handle, by encoding too many different items in different colors. As an example, consider Figure 19-1. It shows population growth versus population size for all 50 US states and the District of Columbia. I have attempted to identify each state by giving it its own color. However, the result is not very useful. Even though we can guess which state is which by looking at the colored points in the plot and in the legend, it takes a lot of effort to go back and forth between the two to try to match them up. There are simply too many different colors, and many of them are quite similar to each other. Even if with a lot of effort we can figure out exactly which state is which, this visualization defeats the purpose of coloring. We should use color to enhance figures and make them easier to read, not to obscure the data by creating visual puzzles.

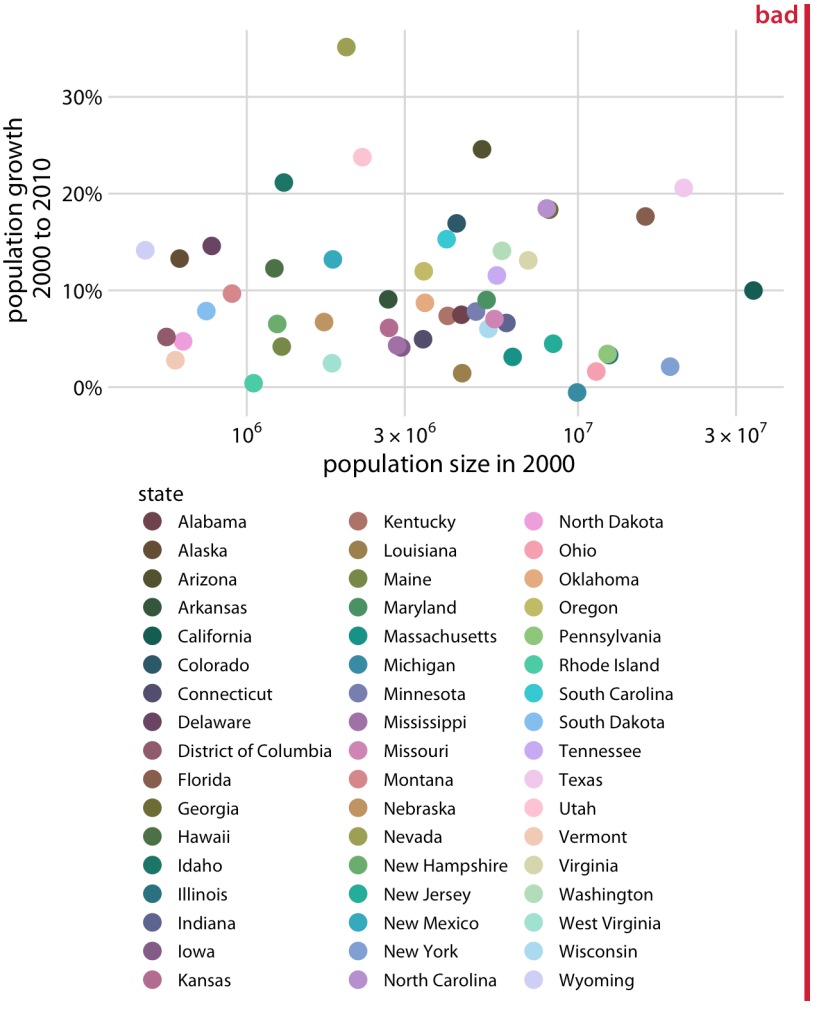


Figure 19-1. Population growth from 2000 to 2010 versus population size in 2000, for all 50 US states and the District of Columbia. Every state is marked in a different color. Because there are so many states, it is very difficult to match the colors in the legend to the dots in the scatterplot. Data source: US Census Bureau.

As a rule of thumb, qualitative color scales work best when there are three to five different categories that need to be colored. Once we reach 8 to 10 different categories or more, the task of matching colors to categories becomes too burdensome to be useful, even if the colors remain sufficiently different to be distinguishable in principle. For the dataset of Figure 19-1, it is probably best to use color only to indicate the geographic region of each state and to identify individual states by direct labeling—i.e., by placing appropriate text labels adjacent to the data points (Figure 19-2). Even

though we cannot label every individual state without making the figure too crowded, direct labeling is the right choice for this figure. In general, for figures such as this one, we don't need to label every single data point. It is sufficient to label a representative subset, for example a set of states we specifically want to call out in the text that will accompany the figure. We always have the option to also provide the underlying data as a table if we want to make sure the reader has access to it in its entirety.

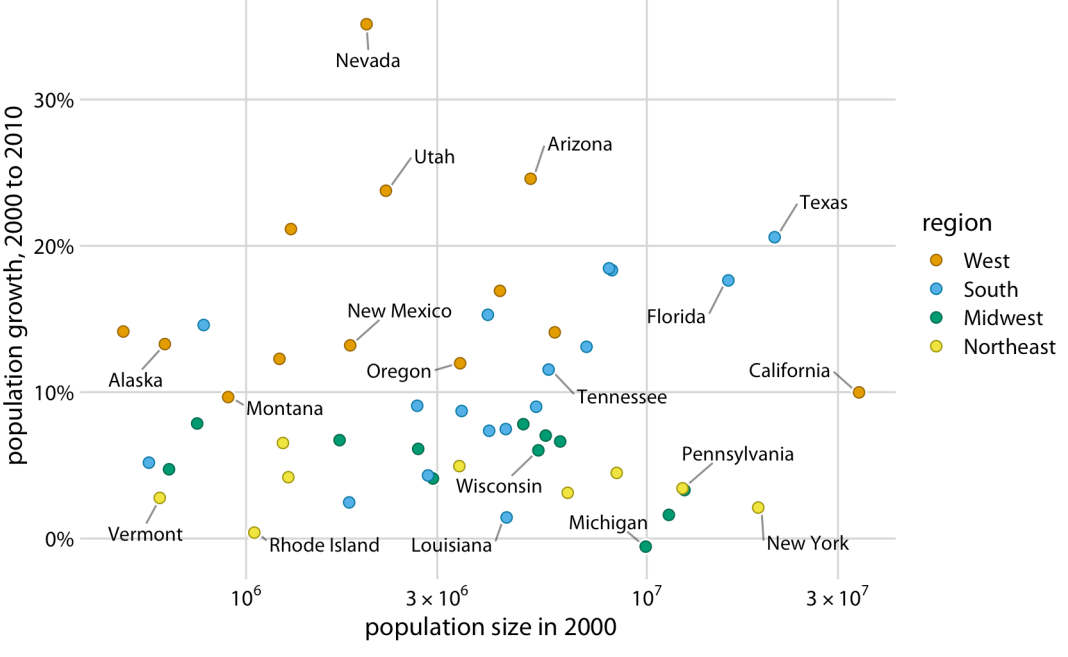


Figure 19-2. Population growth from 2000 to 2010 versus population size in 2000. In contrast to Figure 19-1, I have now colored states by region and have directly labeled a subset of states. The majority of states have been left unlabeled to prevent overcrowding in the figure. Data source: US Census Bureau.



Use direct labeling instead of colors when you need to distinguish between more than about eight categorical items.

A second common problem is coloring for the sake of coloring, without having a clear purpose for the colors. As an example, consider Figure 19-3, which is a variation of Figure 4-2. However, now instead of coloring the bars by geographic regions, I have given each bar its own color, so that in aggregate the bars create a rainbow effect.

This may look like an interesting visual effect, but it is not creating any new insight into the data or making the figure easier to read.

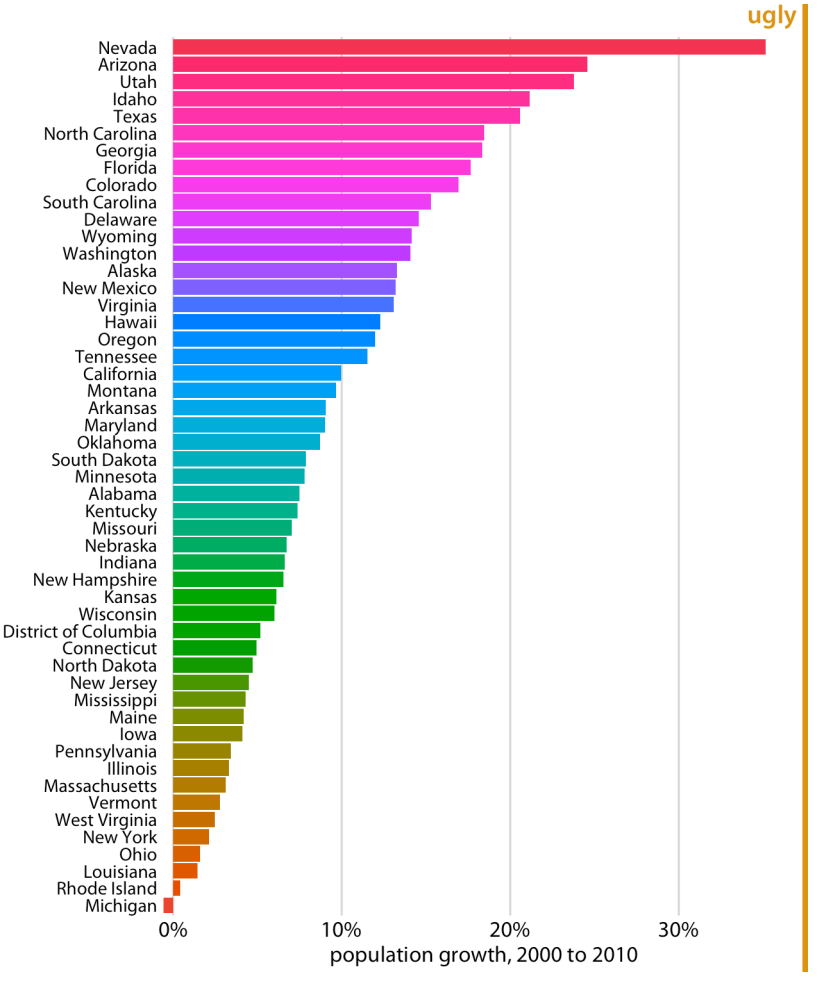


Figure 19-3. Population growth in the US from 2000 to 2010. The rainbow coloring of states serves no purpose and is distracting. Furthermore, the colors are overly saturated. Data source: US Census Bureau.

Besides the gratuitous use of different colors, Figure 19-3 has a second color-related problem: the chosen colors are too saturated and intense. This color intensity makes the figure difficult to look at. For example, it is difficult to read the names of the states without having our eyes drawn to the large, strongly colored areas right next to the state names. Similarly, it is difficult to compare the endpoints of the bars to the underlying grid lines.



Avoid large filled areas of overly saturated colors. They make it difficult for your reader to carefully inspect your figure.

Using Nonmonotonic Color Scales to Encode Data Values

In Chapter 4, I listed two critical conditions for designing sequential color scales that can represent data values: the colors need to clearly indicate which data values are larger or smaller than which other ones, and the differences between colors need to visualize the corresponding differences between data values. Unfortunately, several existing color scales—including very popular ones—violate one or both of these conditions. The most popular such scale is the rainbow scale (Figure 19-4). It runs through all possible colors in the color spectrum. This means the scale is effectively circular; the colors at the beginning and the end are nearly the same (dark red). If these two colors end up next to each other in a plot, we do not instinctively perceive them as representing data values that are maximally apart. In addition, the scale is highly nonmonotonic. It has regions where colors change very slowly and others where colors change rapidly. This lack of monotonicity becomes particularly apparent if we look at the color scale converted to grayscale (Figure 19-4). The scale goes from medium dark to light to very dark and back to medium dark, and there are large stretches where lightness changes very little followed by relatively narrow stretches with large changes in lightness.

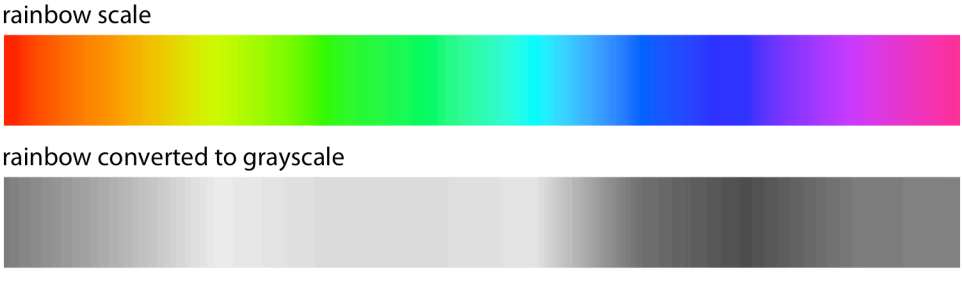


Figure 19-4. The rainbow color scale is highly nonmonotonic. This becomes apparent when the colors are converted to gray values. From left to right, the scale goes from moderately dark to light to very dark and back to moderately dark. In addition, the changes in lightness are nonuniform. The lightest part of the scale (corresponding to the colors yellow, light green, and cyan) takes up almost a third of the entire scale, while the darkest part (corresponding to dark blue) is concentrated in a narrow region of the scale.

In a visualization of actual data, the rainbow scale tends to obscure data features and/or highlight arbitrary aspects of the data (Figure 19-5). As an aside, the colors in

the rainbow scale are also overly saturated. Looking at Figure 19-5 for any extended period of time can be quite uncomfortable.

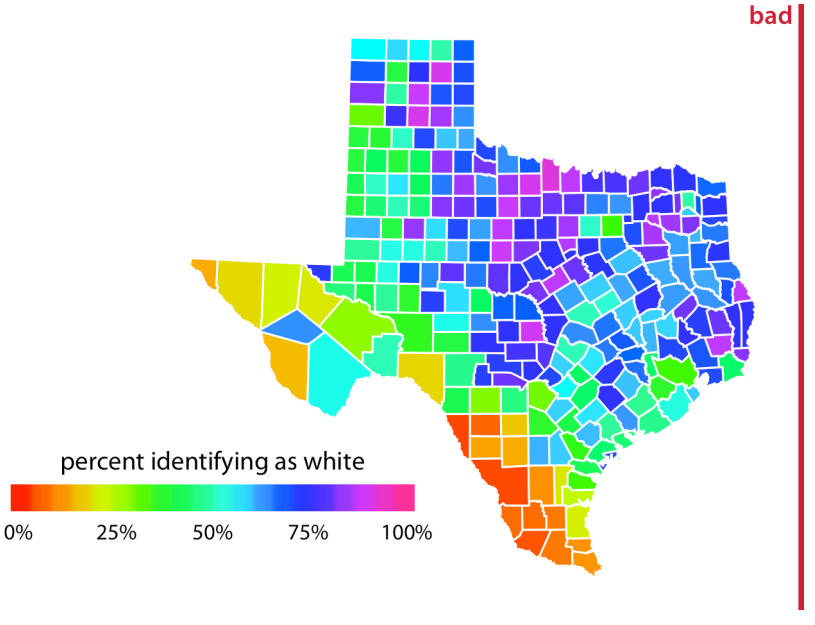


Figure 19-5. Percentage of people identifying as white in Texas counties. The rainbow color scale is not an appropriate scale to visualize continuous data values, because it tends to place emphasis on arbitrary features of the data. Here, it emphasizes counties in which approximately 75% of the population identify as white. Data source: 2010 US Decennial Census.

Not Designing for Color-Vision Deficiency

Whenever we are choosing colors for a visualization, we need to keep in mind that a good proportion of our readers may have some form of color-vision deficiency (i.e., are colorblind). These readers may not be able to distinguish colors that look clearly different to most other people. People with impaired color vision are not literally unable to see any colors, however. Instead, they will typically have difficulty distinguishing certain types of colors, such as red and green (red–green color-vision deficiency) or blue and green (blue–yellow color-vision deficiency). The technical terms for these deficiencies are *deuteranomaly/deuteranopia* and *protanomaly/protanopia* for the red–green variant (where people have difficulty perceiving either green or red, respectively) and *tritanomaly/tritanopia* for the blue–yellow variant (where people have difficulty perceiving blue). The terms ending in "anomaly" refer to some impairment in the perception of the respective color, and the terms ending in "anopia" refer to complete absence of perception of that color. Approximately 8% of males and 0.5%

of females suffer from some sort of color-vision deficiency (CVD); deuteranomaly is the most common form whereas tritanomaly is relatively rare.

As discussed in Chapter 4, there are three fundamental types of color scales used in data visualization: sequential scales, diverging scales, and qualitative scales. Of these three, sequential scales will generally not cause any problems for people with CVD, since a properly designed sequential scale should present a continuous gradient from dark to light colors. Figure 19-6 shows the Heat scale from Figure 4-3 in simulated versions of deuteranomaly, protanomaly, and tritanomaly. While none of these CVD-simulated scales look like the original, they all present a clear gradient from dark to light and they all work well to convey the magnitude of a data value.

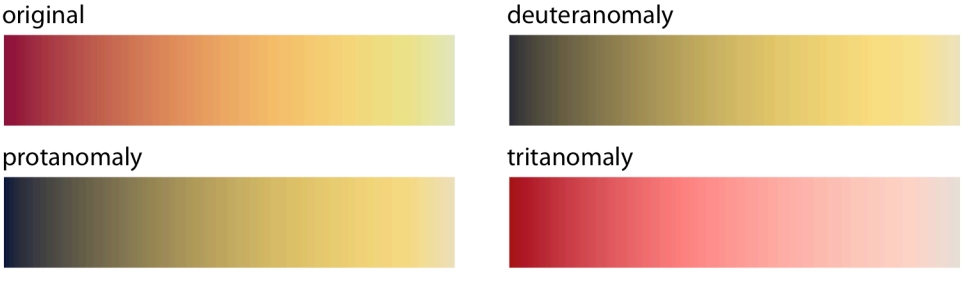


Figure 19-6. Color-vision deficiency simulation of the sequential color scale Heat, which runs from dark red to light yellow. From left to right and top to bottom, we see the original scale and the scale as seen under deuteranomaly, protanomaly, and tritanomaly simulations. Even though the specific colors look different under the three types of CVD, in each case we can see a clear gradient from dark to light. Therefore, this color scale is safe to use for viewers with CVD.

Things become more complicated for diverging scales, because popular color contrasts can be indistinguishable to people with CVD. In particular, the colors red and green provide about the strongest contrast for people with normal color vision but become nearly indistinguishable for deutans (people with deuteranomaly) or protans (people with protanomaly) (Figure 19-7). Similarly, blue-green contrasts are visible for deutans and protans but become indistinguishable for tritans (people with tritanomaly) (Figure 19-8).

These examples might suggest that it is nearly impossible to find two contrasting colors that are safe under all forms of CVD. However, the situation is not that dire. It is often possible to make slight modifications to the colors such that they have the desired character while also being safe for viewers with CVD. For example, the ColorBrewer PiYG (pink to yellow-green) scale from Figure 4-5 looks red–green to people with normal color vision yet remains distinguishable for people with CVD (Figure 19-9).

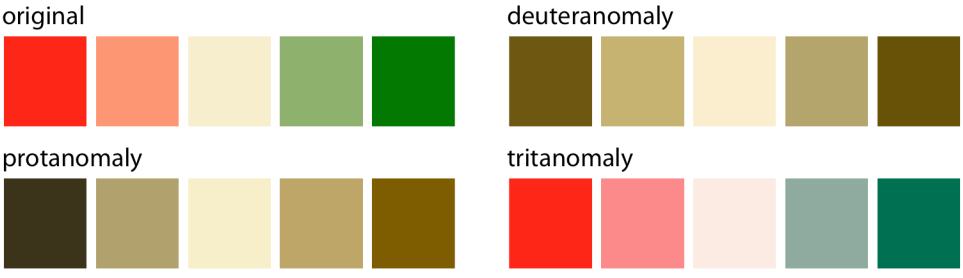


Figure 19-7. A red–green contrast becomes indistinguishable under red–green CVD (deuteranomaly or protanomaly).

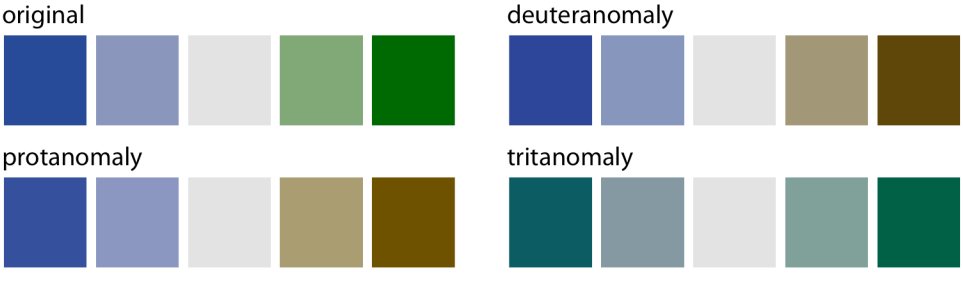


Figure 19-8. A blue–green contrast becomes indistinguishable under blue–yellow CVD (tritanomaly).



Figure 19-9. The ColorBrewer PiYG (pink to yellow-green) scale from Figure 4-5 looks like a red–green contrast to people with regular color vision but works for people with all forms of color-vision deficiency. It works because the reddish color is actually pink (a mix of red and blue) while the greenish color also contains yellow. The difference in the blue component between the two colors can be picked up by deutans or protans, and the difference in the red component can be picked up by tritans.

Things are most complicated for qualitative scales, because there we need many different colors and they all need to be distinguishable from each other under all forms

of CVD. My preferred qualitative color scale, which I use extensively throughout this book, was developed specifically to address this challenge (Figure 19-10). By providing eight different colors, the palette works for nearly any scenario with discrete colors. As discussed at the beginning of this chapter, you should probably not color-code more than eight different items in a plot anyway.



Figure 19-10. Qualitative color palette for all color-vision deficiencies [Okabe and Ito 2008]. The alphanumeric codes represent the colors in RGB space, encoded as hexadecimals. In many plot libraries and image manipulation programs, you can just enter these codes directly. If your software does not take hexadecimals directly, you can also use the values in Table 19-1.

Table 19-1. Colorblind-friendly color scale [Okabe and Ito 2008].

Name	Hex code	Hue	C, M, Y, K (%)	R, G, B (0–255)	R, G, B (%)
Orange	#E69F00	41°	0, 50, 100, 0	230, 159, 0	90, 60, 0
Sky blue	#56B4E9	202°	80, 0, 0, 0	86, 180, 233	35, 70, 90
Bluish green	#009E73	164°	97, 0, 75, 0	0, 158, 115	0, 60, 50
Yellow	#F0E442	56°	10, 5, 90, 0	240, 228, 66	95, 90, 25
Blue	#0072B2	202°	100, 50, 0, 0	0, 114, 178	0, 45, 70
Vermilion	#D55E00	27°	0, 80, 100, 0	213, 94, 0	80, 40, 0
Reddish purple	#CC79A7	326°	10, 70, 0, 0	204, 121, 167	80, 60, 70
Black	#000000	N/A	0, 0, 0, 100	0, 0, 0	0, 0, 0

While there are several good, CVD-safe color scales readily available, we need to recognize that they are not magic bullets. It is very possible to use a CVD-safe scale and yet produce a figure a person with CVD cannot decipher. One critical parameter is the size of the colored graphical elements. Colors are much easier to distinguish when they are applied to large areas than to small ones or thin lines [Stone, Albers Szafir, and Setlur 2014], and this effect is exacerbated under CVD (Figure 19-11). In addition to the various color design considerations discussed in this chapter and in Chapter 4, I recommend to view color figures under CVD simulations to get a sense of what they may look like for a person with CVD. There are several online services and desktop apps available that allow you to run arbitrary figures through a CVD simulation.

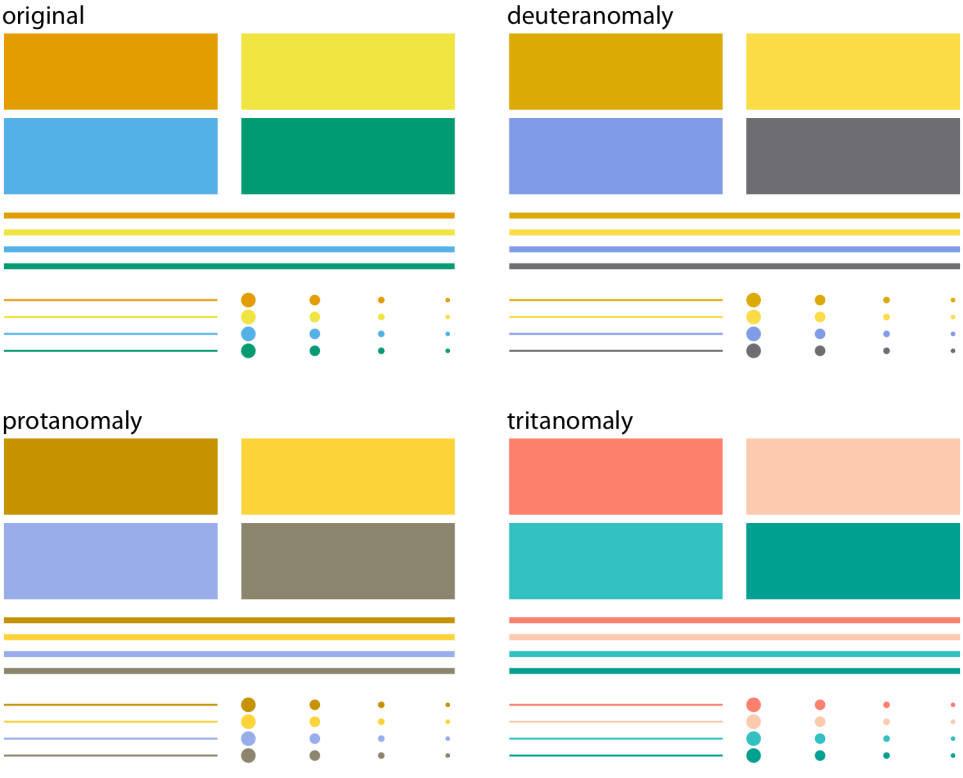


Figure 19-11. Colored elements become difficult to distinguish at small sizes. The top-left panel (labeled "original") shows four rectangles, four thick lines, four thin lines, and four groups of points, all colored in the same four colors. We can see that the colors become more difficult to distinguish the smaller or thinner the visual elements are. This problem becomes exacerbated in the CVD simulations, where the colors are already more difficult to distinguish even for the large graphical elements.



To make sure your figures work for people with CVD, don't just rely on specific color scales. Instead, test your figures in a CVD simulator.

CHAPTER 20

Redundant Coding

In Chapter 19, we saw that color cannot always convey information as effectively as we might wish. If we have many different items we want to identify, doing so by color may not work. It will be difficult to match the colors in the plot to the colors in the legend (Figure 19-1). And even if we only need to distinguish two or three different items, color may fail if the colored items are very small (Figure 19-11) and/or the colors look similar for people suffering from color-vision deficiency (Figures 19-7 and 19-8). The general solution in all these scenarios is to use color to enhance the visual appearance of the figure without relying entirely on color to convey key information. I refer to this design principle as *redundant coding*, because it prompts us to encode data redundantly, using multiple different aesthetic dimensions.

Designing Legends with Redundant Coding

Scatterplots of several groups of data are frequently designed such that the points representing different groups differ only in their color. As an example, consider Figure 20-1, which shows the sepal width versus the sepal length of three different *Iris* species. (Sepals are the outer leaves of flowers in flowering plants.) The points representing the different species differ in their colors, but otherwise all points look exactly the same. Even though this figure contains only three distinct groups of points, it is difficult to read even for people with normal color vision. The problem arises because the data points for the two species *Iris virginica* and *Iris versicolor* intermingle, and their two respective colors, green and blue, are not particularly distinct from each other.

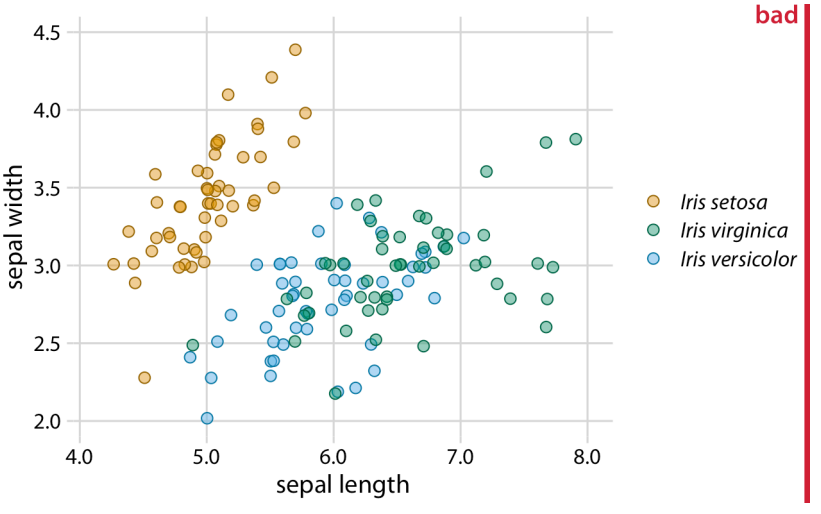


Figure 20-1. Sepal width versus sepal length for three different Iris *species (*Iris setosa*,* Iris virginica*, and* Iris versicolor*). Each point represents the measurements for one plant sample. A small amount of jitter has been applied to all point positions to prevent overplotting. The figure is labeled "bad" because the* virginica *points in green and the* versicolor *points in blue are difficult to distinguish from each other. Data source: [Fisher 1936].*

Surprisingly, the green and blue points look more distinct for people with red–green color-vision deficiency (deuteranomaly or protanomaly) than for people with normal color vision (compare Figure 20-2, top row, to Figure 20-1). On the other hand, for people with blue–yellow deficiency (tritanomaly), the blue and green points look very similar (Figure 20-2, bottom left). And if we print out the figure in grayscale (i.e., we *desaturate* the figure), we cannot distinguish any of the *Iris* species (Figure 20-2, bottom right).

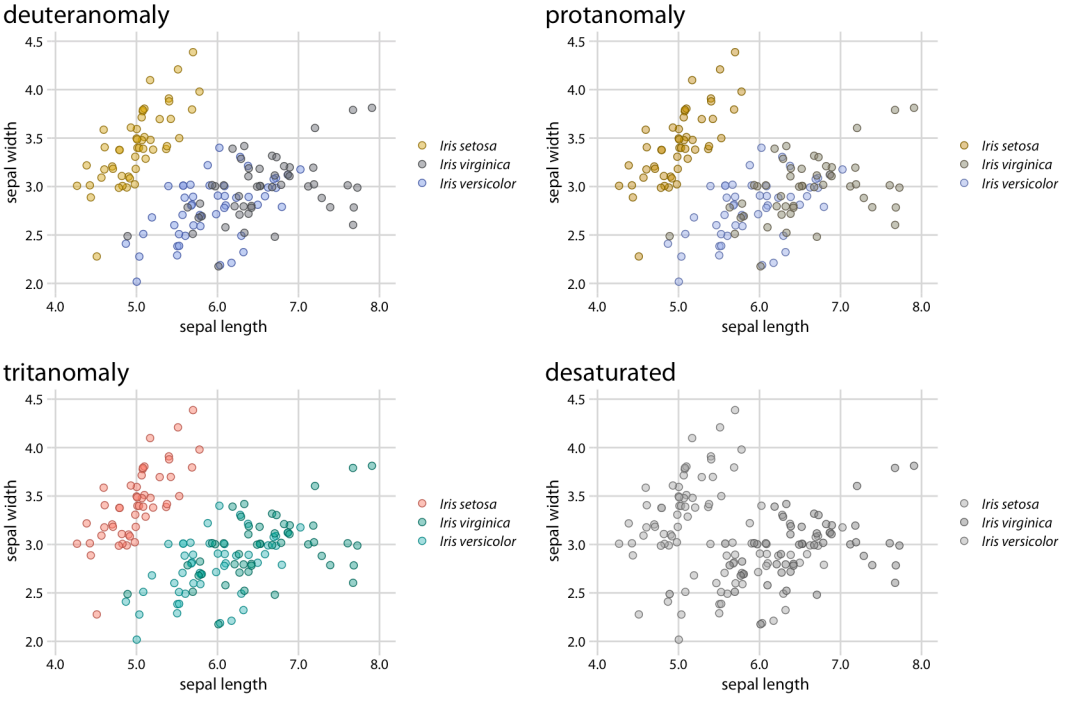


Figure 20-2. Color-vision deficiency simulation of Figure 20-1. Data source: [Fisher 1936].

There are two simple improvements we can make to Figure 20-1 to alleviate these issues. First, we can swap the colors used for *Iris setosa* and *Iris versicolor*, so that the blue is no longer directly next to the green (Figure 20-3). Second, we can use three different symbol shapes, so that the points all look different. With these two changes, both the original version of the figure (Figure 20-3) and the versions under color-vision deficiency and in grayscale (Figure 20-4) become legible.

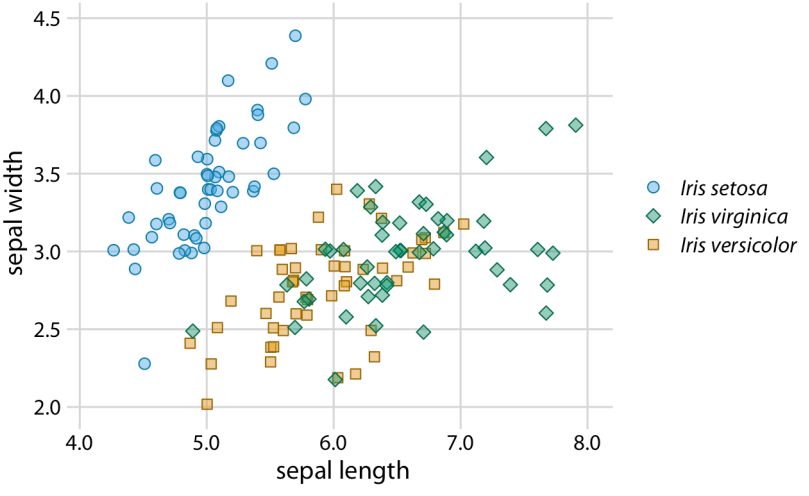


Figure 20-3. Sepal width versus sepal length for three different Iris *species. Compared to Figure 20-1, we have swapped the colors for* Iris setosa *and* Iris versicolor *and we have given each* Iris *species its own point shape. Data source: [Fisher 1936].*

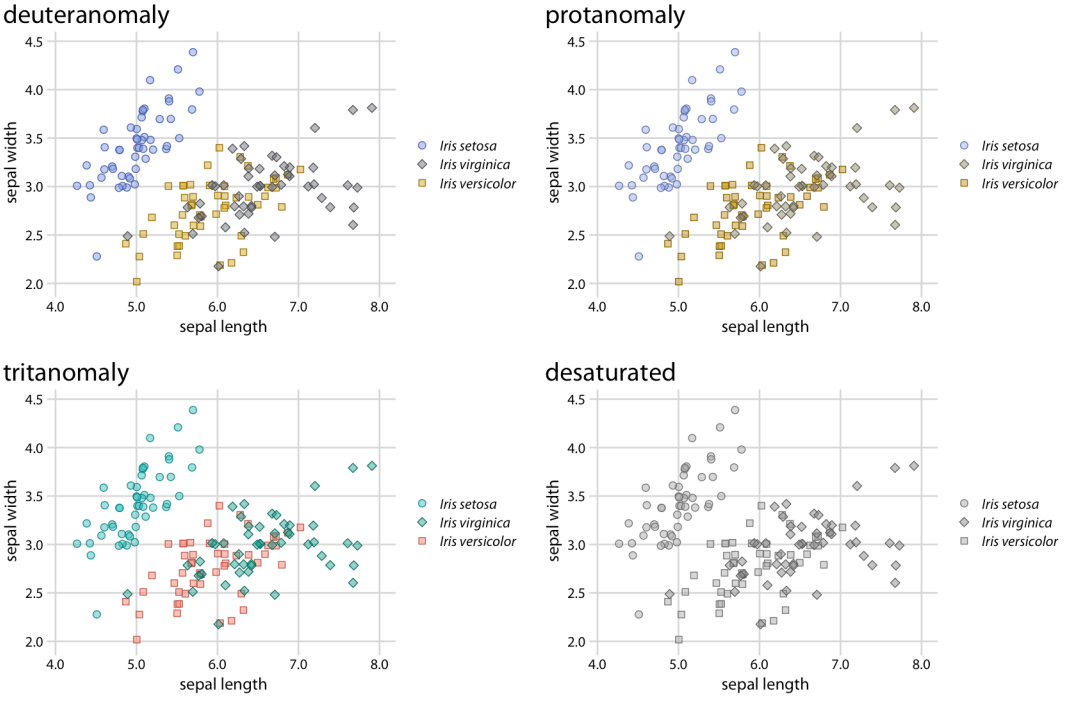


Figure 20-4. Color-vision deficiency simulation of Figure 20-3. Because of the use of different point shapes, even the fully desaturated grayscale version of the figure is legible. Data source: [Fisher 1936].

Changing the point shape is a simple strategy for scatterplots, but it doesn't necessarily work for other types of plots. In line plots, we could change the line type (solid, dashed, dotted, etc.; see also Figure 2-1), but using dashed or dotted lines often yields sub-optimal results. In particular, dashed or dotted lines usually don't look good unless they are perfectly straight or only gently curved, and in either case they create visual noise. Also, it frequently requires significant mental effort to match different types of dash or dot–dash patterns from the plot to the legend. So what do we do with a visualization such as Figure 20-5, which uses lines to show the change in stock price over time for four different major tech companies?

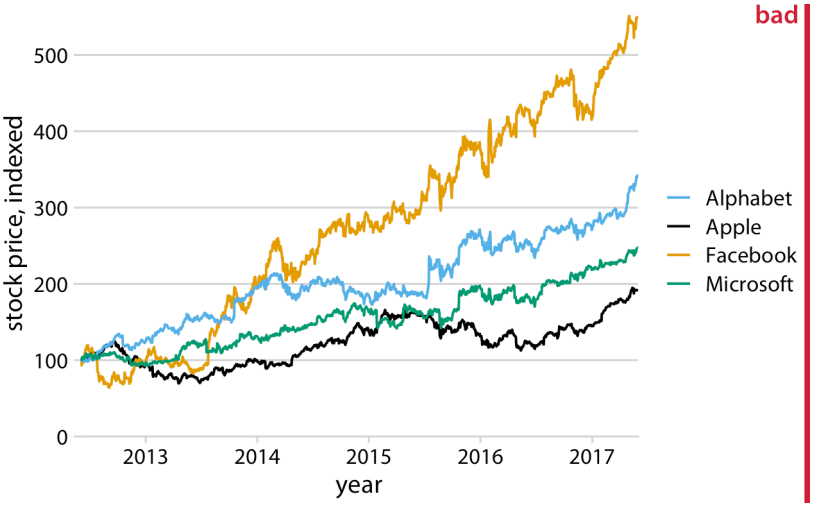


Figure 20-5. Stock price over time for four major tech companies. The stock price for each company has been normalized to equal 100 in June 2012. This figure is labeled as "bad" because it takes considerable mental energy to match the company names in the legend to the data curves. Data source: Yahoo! Finance.

The figure contains four lines representing the stock prices of the four different companies. The lines are color-coded using a colorblind-friendly color scale. Thus, it should be relatively straightforward to associate each line with the corresponding company—yet it is not. The problem here is that the data lines have a visual order. The yellow line, representing Facebook, is perceived as the highest line, and the black line, representing Apple, is perceived as the lowest, with Alphabet and Microsoft in between, in that order. Yet the order of the four companies in the legend is Alphabet, Apple, Facebook, Microsoft (alphabetical order). Thus, the perceived order of the data lines differs from the order of the companies in the legend, and it takes a surprising amount of mental effort to match data lines with company names.

This problem arises commonly with plotting software that autogenerates legends. The plotting software has no concept of the visual order the viewer will perceive. Instead, the software sorts the legend by some other order, most commonly alphabetical. We can fix this problem by manually reordering the entries in the legend so they match the perceived ordering in the data (Figure 20-6). The result is a figure that makes it much easier to match the legend to the data.

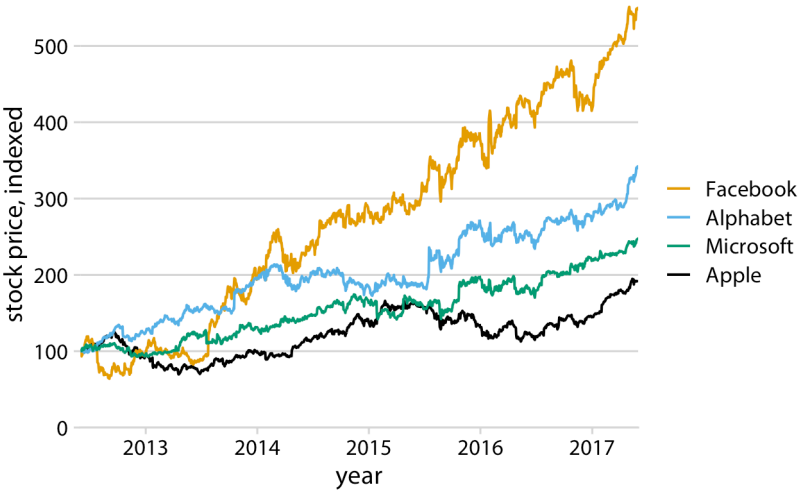


Figure 20-6. Stock price over time for four major tech companies. Compared to Figure 20-5, the entries in the legend have now been ordered such that they match the perceived visual order of the data lines, with Facebook the highest and Apple the lowest. Data source: Yahoo! Finance.



If there is a visual ordering in your data, make sure to match it in the legend.

Matching the legend order to the data order is always helpful, but the benefits are particularly obvious under color-vision deficiency simulation (Figure 20-7). For example, it helps in the tritanomaly version of the figure, where the blue and the green become difficult to distinguish (Figure 20-7, bottom left). It also helps in the grayscale version (Figure 20-7, bottom right). Even though the two colors for Facebook and Alphabet have virtually the same gray value, we can see that Microsoft and Apple are represented by darker colors and take the bottom two spots. Therefore, we correctly assume that the highest line corresponds to Facebook and the second-highest line to Alphabet.

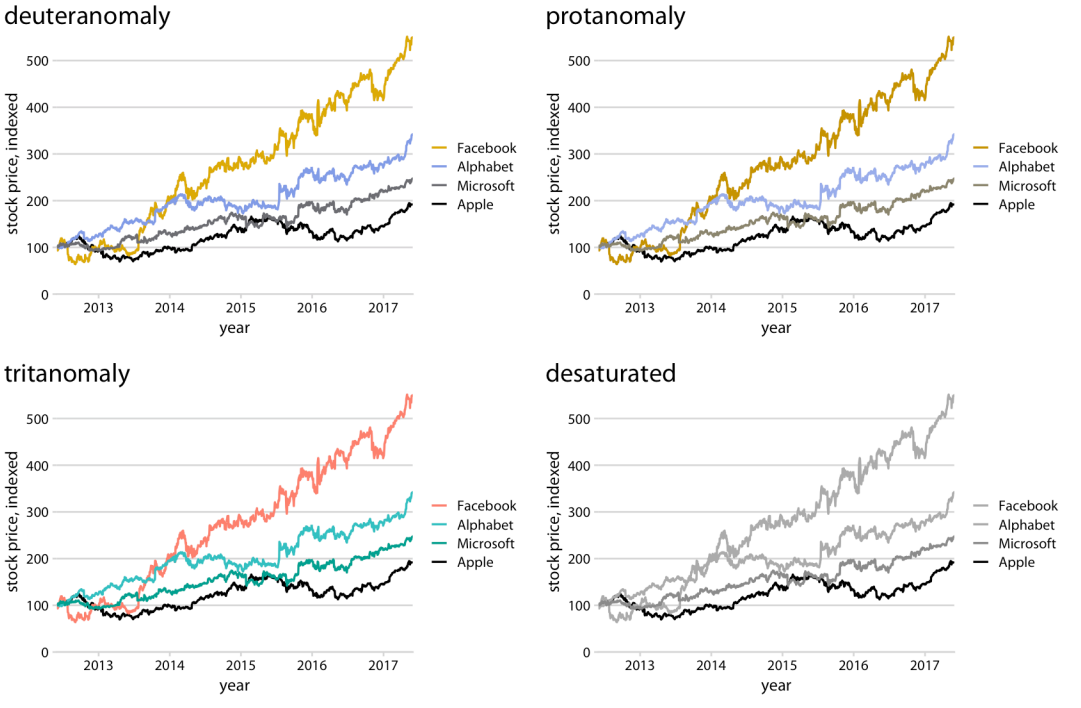


Figure 20-7. Color-vision deficiency simulation of Figure 20-6. Data source: Yahoo! Finance.

Designing Figures Without Legends

Even though legend legibility can be improved by encoding data redundantly, in multiple aesthetics, legends always put an extra mental burden on the reader. In reading a legend, the reader needs to pick up information in one part of the visualization and then transfer it over to a different part. We can typically make our readers' lives easier if we eliminate the legend altogether. Eliminating the legend does not mean, however, that we simply don't provide one and instead write sentences such as "The yellow dots represent *Iris versicolor*" in the figure caption. Eliminating the legend means that we design the figure in such a way that it is immediately obvious what the various graphical elements represent, even if no explicit legend is present.

The general strategy we can employ is called *direct labeling*, whereby we incorporate appropriate text labels or other visual elements that serve as guideposts to the rest of the figure. We have previously encountered direct labeling in Chapter 19 (Figure 19-2), as an alternative to drawing a legend with over 50 distinct colors. To apply the direct labeling concept to the stock price figure, we place the name of each company right next to the end of its respective data line (Figure 20-8).

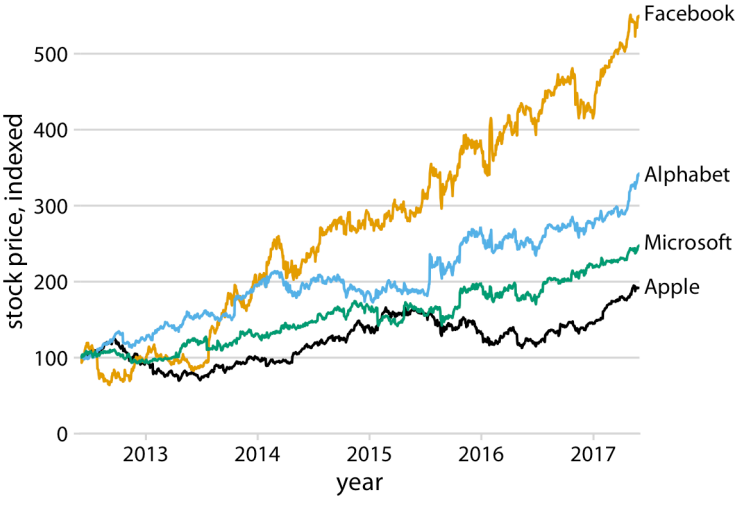


Figure 20-8. Stock price over time for four major tech companies. The stock price for each company has been normalized to equal 100 in June 2012. Data source: Yahoo! Finance.



Whenever possible, design your figures so they don't need a separate legend.

We can also apply the direct labeling concept to the *Iris* data from the beginning of this chapter, specifically Figure 20-3. Because it is a scatterplot of many points that separate into three different groups, we need to directly label the groups rather than the individual points. One solution is to draw ellipses that enclose the majority of the points and then label the ellipses (Figure 20-9).

For density plots, we can similarly direct-label the curves rather than providing a color-coded legend (Figure 20-10). In both Figures 20-9 and 20-10, I have colored the text labels in the same colors as the data. Colored labels can greatly enhance the direct labeling effect, but they can also turn out poorly. If the text labels are printed in a color that is too light, then the labels become difficult to read. And because text consists of very thin lines, colored text often appears to be lighter than an adjacent filled area of the same color. I generally circumvent these issues by using two different shades of each color, a light one for filled areas and a dark one for lines, outlines, and text. If you carefully inspect Figure 20-9 or 20-10, you will see how each data point or shaded area is filled with a light color and has an outline drawn in a darker color of the same hue. The text labels are drawn in the same darker colors.

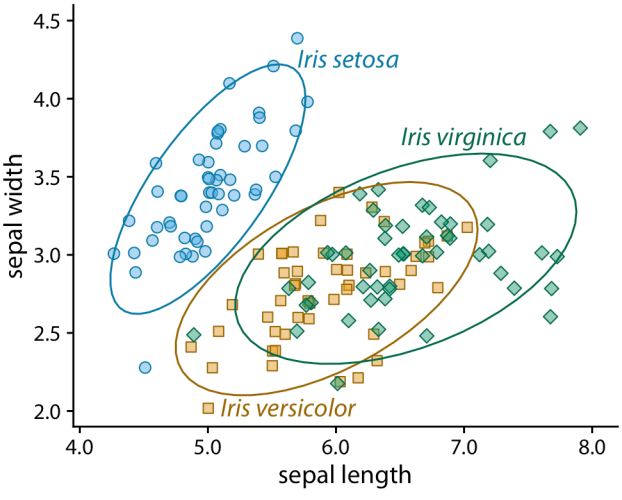


Figure 20-9. Sepal width versus sepal length for three different Iris *species. The points representing different* Iris *species have been directly labeled with colored ellipses and text labels. Compared to Figure 20-3, I have removed the background grid here because the figure was becoming too busy. Data source: [Fisher 1936].*

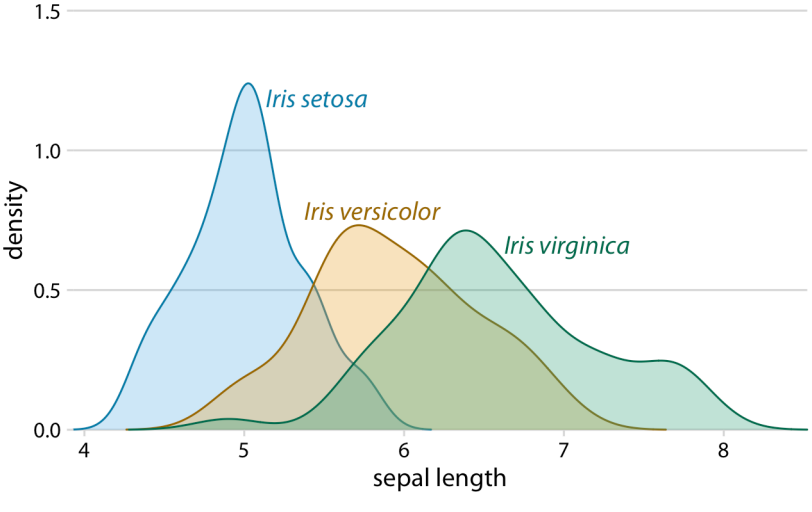


Figure 20-10. Density estimates of the sepal lengths of three different Iris *species. Each density estimate is directly labeled with the respective species name. Data source: [Fisher 1936].*

We can also use density plots such as the one in Figure 20-10 as a legend replacement, by placing the density plots into the margins of a scatterplot (Figure 20-11). This

allows us to direct-label the marginal density plots rather than the central scatterplot and hence results in a figure that is somewhat less cluttered than Figure 20-9 with its directly labeled ellipses.

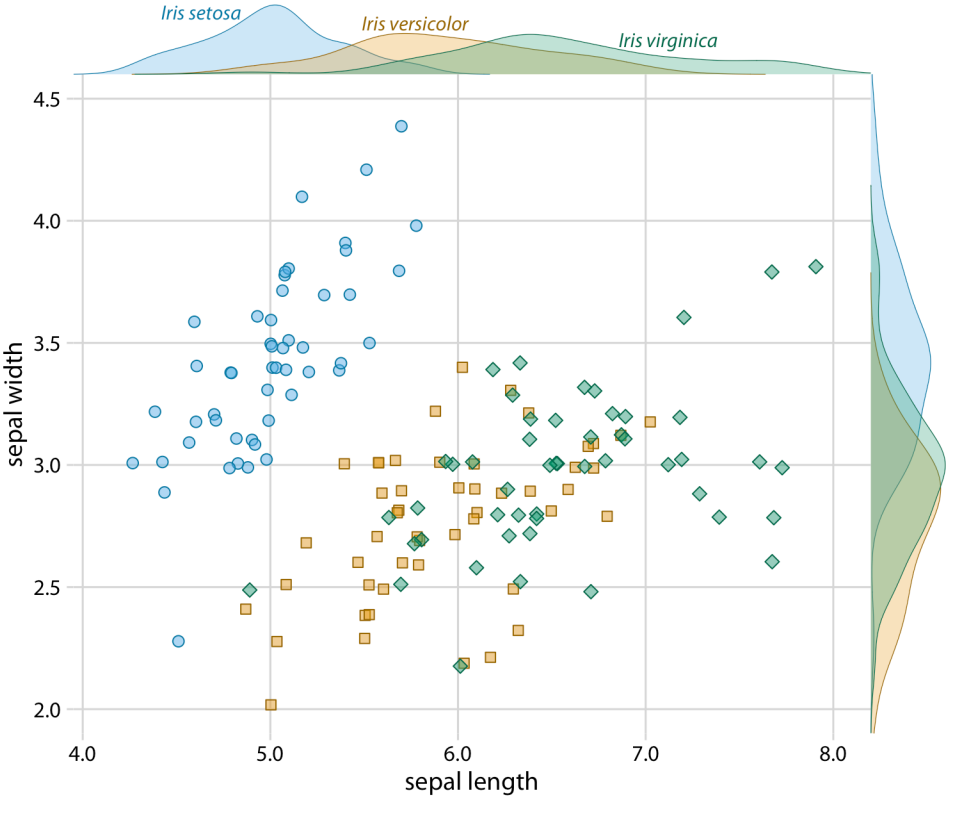


Figure 20-11. Sepal width versus sepal length for three different Iris *species, with marginal density estimates of each variable for each species. Data source: [Fisher 1936].*

And finally, whenever we encode a single variable in multiple aesthetics, we don't normally want multiple separate legends for the different aesthetics. Instead, there should be a single legend-like visual element that conveys all the mappings at once. In the case where we map the same variable onto a position along a major axis and onto color, this implies that the reference color bar should run along and be integrated into the same axis. Figure 20-12 shows a case where we map temperature to both a position along the *x* axis and to color, and where we therefore have integrated the color legend into the *x* axis.

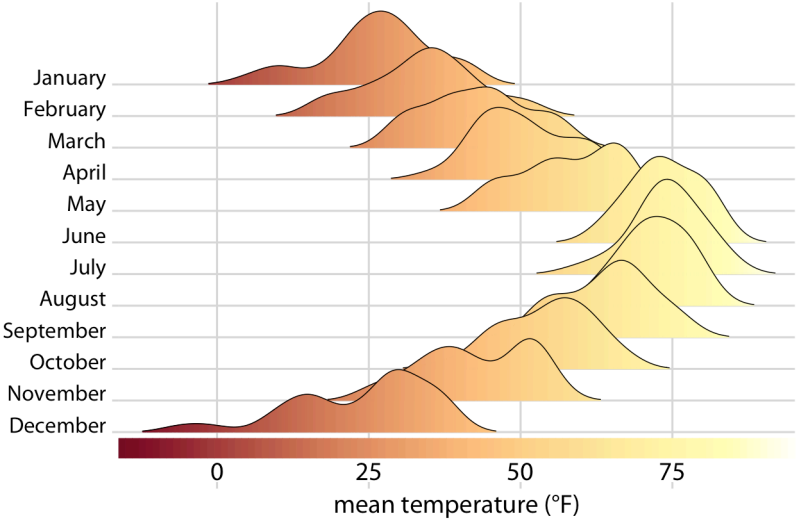


Figure 20-12. Temperatures in Lincoln, NE, in 2016. This figure is a variation of Figure 9-9. Temperature is now shown both by location along the x *axis and by color, and a color bar along the* x *axis visualizes the scale that converts temperatures into colors. Data source: Weather Underground.*

CHAPTER 23

Balance the Data and the Context

We can broadly subdivide the graphical elements in any visualization into elements that represent data and elements that do not. The former are elements such as the points in a scatterplot, the bars in a histogram or bar plot, or the shaded areas in a heatmap. The latter are elements such as plot axes, axis ticks and labels, axis titles, legends, and plot annotations. These elements generally provide context for the data and/or visual structure to the plot. When designing a plot, it can be helpful to think about the amount of ink (Chapter 17) used to represent the data and context. A common recommendation is to reduce the amount of non-data ink, and following this advice can often yield less cluttered and more elegant visualizations. At the same time, context and visual structure are important, and overly minimizing the plot elements that provide them can result in figures that are difficult to read, confusing, or simply not that compelling.

Providing the Appropriate Amount of Context

The idea that distinguishing between data and non-data ink may be useful was popularized by Edward Tufte in his book *The Visual Display of Quantitative Information* [Tufte 2001]. Tufte introduces the concept of the "data–ink ratio," which he defines as the "proportion of a graphic's ink devoted to the non-redundant display of data information." He then writes (emphasis mine):

> Maximize the data–ink ratio, *within reason.*

I have emphasized the phrase "within reason" because it is critical and frequently forgotten. In fact, I think that Tufte himself forgets it in the remainder of his book, where he advocates overly minimalistic designs that, in my opinion, are neither elegant nor easy to decipher. If we interpret the phrase "maximize the data–ink ratio" to mean "remove clutter and strive for clean and elegant designs," then I think it is reasonable advice. But if we interpret it as "do everything you can to remove non-data

ink," then it will result in poor design choices. If we go too far in either direction we will end up with ugly figures. However, away from the extremes there is a wide range of designs that are all acceptable and may be appropriate in different settings.

To explore the extremes, let's consider a figure that has far too much non-data ink (Figure 23-1). The colored points in the plot panel (the framed center area containing data points) are data ink. Everything else is non-data ink. The non-data ink includes a frame around the entire figure, a frame around the plot panel, and a frame around the legend. None of these frames are needed. We also see a prominent and dense background grid that draws attention away from the actual data points. By removing the frames and minor grid lines and by drawing the major grid lines in a light gray, we arrive at Figure 23-2. In this version of the figure, the actual data points stand out much more clearly, and they are perceived as the most important component of the figure.

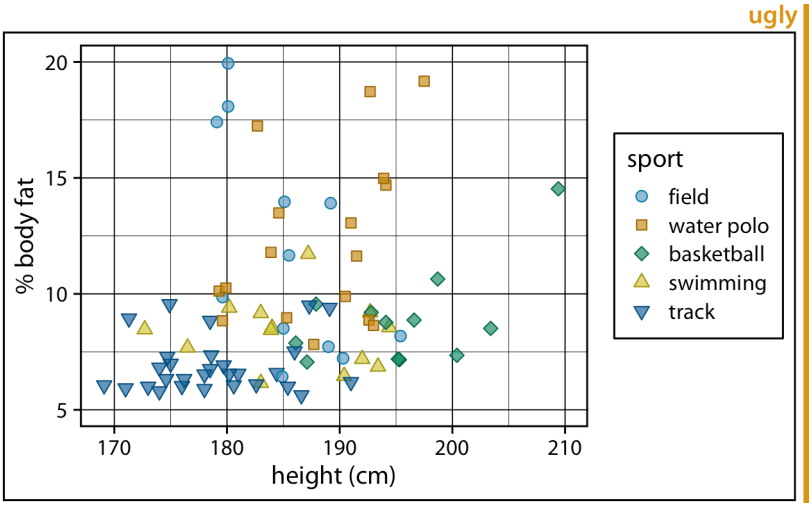


Figure 23-1. Percent body fat versus height in professional male Australian athletes. Each point represents one athlete. This figure devotes way too much ink to non-data. There are unnecessary frames around the entire figure, around the plot panel, and around the legend. The coordinate grid is very prominent, and its presence draws attention away from the data points. Data source: [Telford and Cunningham 1991].

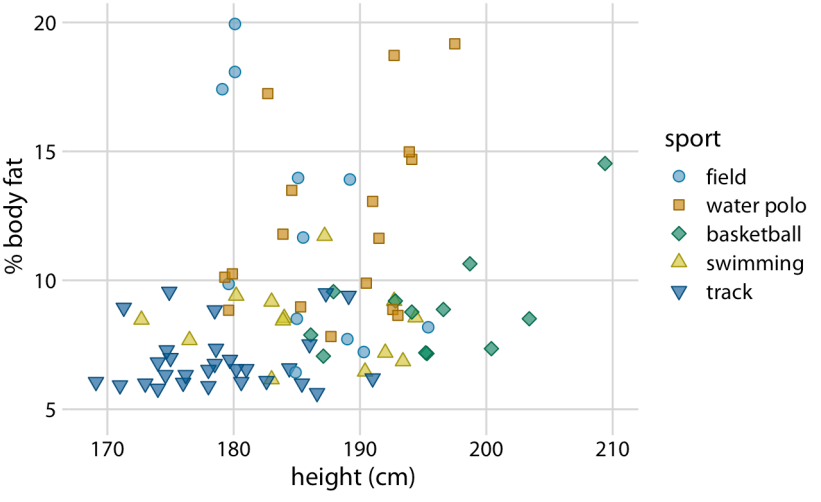


Figure 23-2. Percent body fat versus height in professional male Australian athletes. This figure is a cleaned-up version of Figure 23-1. Unnecessary frames have been removed, minor grid lines have been removed, and major grid lines have been drawn in light gray to stand back relative to the data points. Data source: [Telford and Cunningham 1991].

At the other extreme, we might end up with a figure such as Figure 23-3, which is a minimalist version of Figure 23-2. In this figure, the axis tick labels and titles have been made so faint that they are hard to see. If we just glance at the figure we will not immediately perceive what data is actually shown. We only see points floating in space. Moreover, the legend annotations are so faint that the points in the legend could be mistaken for data points. This effect is amplified because there is no visual separation between the plot area and the legend. Notice how the background grid in Figure 23-2 both anchors the points in space and sets off the data area from the legend area. Both of these effects have been lost in Figure 23-3.

In Figure 23-2, I am using an open background grid and no axis lines or frame around the plot panel. I like this design because it conveys to the viewer that the range of possible data values extends beyond the axis limits. For example, even though Figure 23-2 shows no athlete taller than 210 cm, such an athlete could conceivably exist. However, some authors prefer to delineate the extent of the plot panel, by drawing a frame around it (Figure 23-4). Both options are reasonable, and which is preferable is primarily a matter of personal opinion. One advantage of the framed version is that it visually separates the legend from the plot panel.

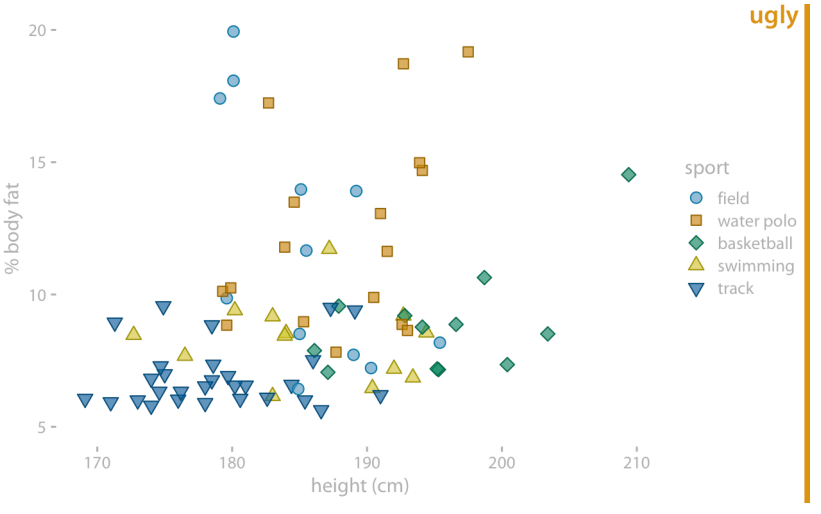


Figure 23-3. Percent body fat versus height in professional male Australian athletes. In this example, the concept of removing non-data ink has been taken too far. The axis tick labels and title are too faint and are barely visible. The data points seem to float in space. The points in the legend are not sufficiently set off from the data points, and the casual observer might think they are part of the data. Data source: [Telford and Cunningham 1991].

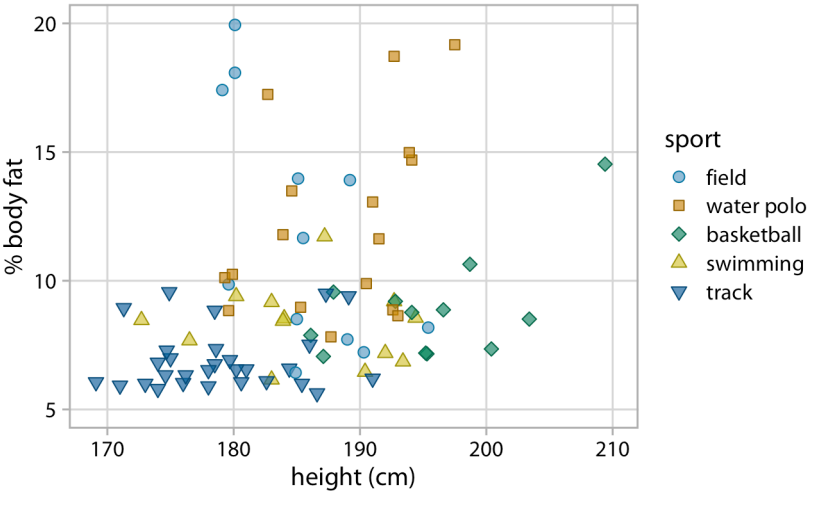


Figure 23-4. Percent body fat versus height in professional male Australian athletes. This figure adds a frame around the plot panel of Figure 23-2, and this frame helps separate the legend from the data. Data source: [Telford and Cunningham 1991].

Figures with too little non-data ink commonly suffer from the effect that figure elements appear to float in space, without clear connection or reference to anything. This problem tends to be particularly severe in small multiples plots. Figure 23-5 shows a small multiples plot comparing six different bar plots, but it looks more like a piece of modern art than a useful data visualization. The bars are not anchored to a baseline and the individual plot facets are not clearly delineated. We can resolve these issues by adding a light gray background and thin horizontal grid lines to each facet (Figure 23-6).



Figure 23-5. Survival of passengers on the Titanic*, broken down by gender and class. This small multiples plot is too minimalistic. The individual facets are not framed, so it's difficult to see which part of the figure belongs to which facet. Further, the individual bars are not anchored to a baseline, and they seem to float. Data source: Encyclopedia Titanica.*

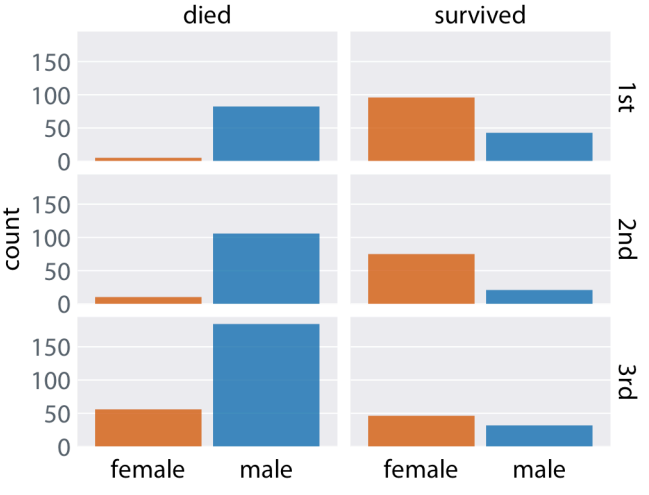


Figure 23-6. Survival of passengers on the Titanic*, broken down by gender and class. This is an improved version of Figure 23-5. The gray background in each facet clearly delineates the six groupings (survived or died in 1st, 2nd, or 3rd class) that make up this plot. Thin horizontal lines in the background provide a reference for the bar heights and facilitate comparison of bar heights among facets. Alternatively, we could put a frame around each individual plot panel and use gray bars to highlight the grouping variables (see Figure 21-1). Data source: Encyclopedia Titanica.*

Background Grids

Grid lines in the background of a plot can help the reader discern specific data values and compare values in one part of a plot to values in another part. At the same time, grid lines can add visual noise, in particular when they are prominent or densely spaced. Reasonable people can disagree about whether to use a grid or not, and if so how to format it and how densely to space it. Throughout this book I am using a variety of different grid styles, to highlight that there isn't necessarily one best choice.

The R software ggplot2 has popularized a style using a fairly prominent background grid of white lines on a gray background. Figure 23-7 shows an example in this style. The figure displays the change in stock price of four major tech companies over a five-year window, from 2012 to 2017. With apologies to the ggplot2 author Hadley Wickham, for whom I have the utmost respect, I don't find the white-on-gray background grid particularly attractive. To my eye, the gray background can detract from the actual data, and a grid with major and minor lines can be too dense. I also find the gray squares in the legend confusing.

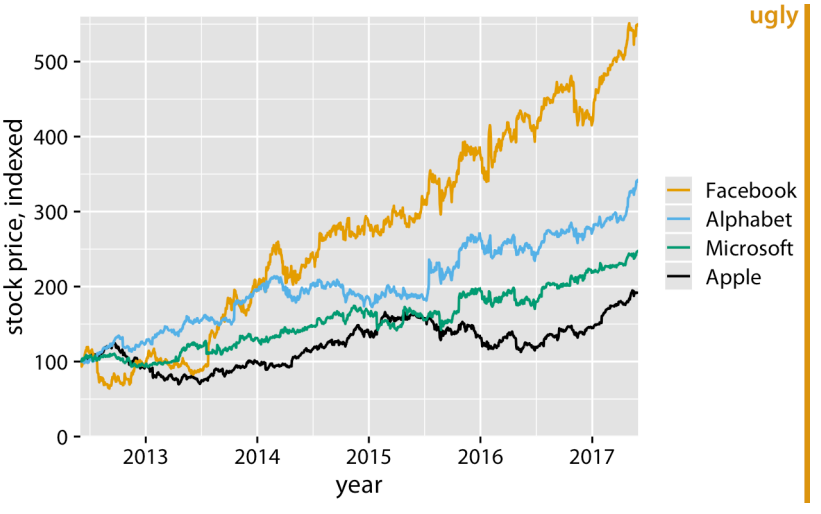


Figure 23-7. Stock price over time for four major tech companies. The stock price for each company has been normalized to equal 100 in June 2012. This figure mimics the ggplot2 default look, with white major and minor grid lines on a gray background. In this particular example, I think the grid lines overpower the data lines, and the result is a figure that is not well balanced and that doesn't place sufficient emphasis on the data. Data source: Yahoo! Finance.

Arguments in favor of the gray background include that it both helps the plot to be perceived as a single visual entity and prevents the plot from appearing as a white box in surrounding dark text [Wickham 2016]. I completely agree with the first point, and it was the reason I used gray backgrounds in Figure 23-6. For the second point, I'd like to caution that the perceived darkness of text will depend on the font size, font face, and line spacing, and the perceived darkness of a figure will depend on the absolute amount and color of ink used, including all data ink. A scientific paper typeset in dense, 10-point Times New Roman will look much darker than a coffee table book typeset in 14-point Palatino with one-and-a-half line spacing. Likewise, a scatterplot of 5 data points in yellow will look much lighter than a scatterplot of 10,000 data points in black. If you want to use a gray figure background, consider the color intensity of your figure foreground, as well as the expected layout and typography of the text around your figures, and adjust the choice of your background gray accordingly. Otherwise, it could happen that your figures end up standing out as dark boxes among the surrounding lighter text. Also, keep in mind that the colors you use to plot your data need to work with the gray background. We tend to perceive colors differently against different backgrounds, and a gray background requires darker and more saturated foreground colors than a white background.

We can go all the way in the opposite direction and remove both the background and the grid lines (Figure 23-8). In this case, we need visible axis lines to frame the plot and keep it as a single visual unit. For this particular figure, I think this choice is a worse option, and I have labeled it as "bad." In the absence of any background grid whatsoever, the curves seem to float in space, and it's difficult to reference the final values on the right to the axis ticks on the left.

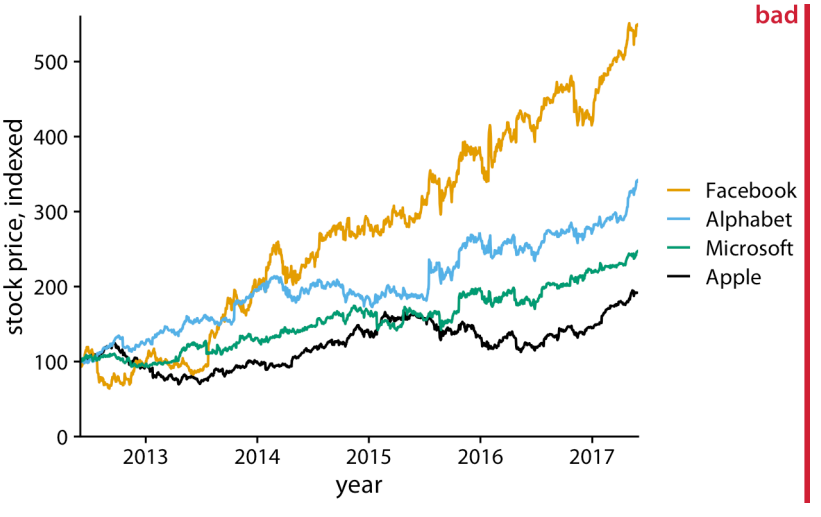


Figure 23-8. Indexed stock price over time for four major tech companies. In this variant of Figure 23-7, the data lines are not sufficiently anchored. This makes it difficult to ascertain to what extent they have deviated from the index value of 100 at the end of the covered time interval. Data source: Yahoo! Finance.

At the absolute minimum, we need to add one horizontal reference line. Since the stock prices in Figure 23-8 are indexed to 100 in June 2012, marking this value with a thin horizontal line at $y = 100$ helps a lot (Figure 23-9). Alternatively, we can use a minimal "grid" of horizontal lines. For a plot where we are primarily interested in the change in y values, vertical grid lines are not needed. Moreover, grid lines positioned at only the major axis ticks will often be sufficient, and the axis line can be omitted or made very thin since the horizontal lines mark the extent of the plot (Figure 23-10).

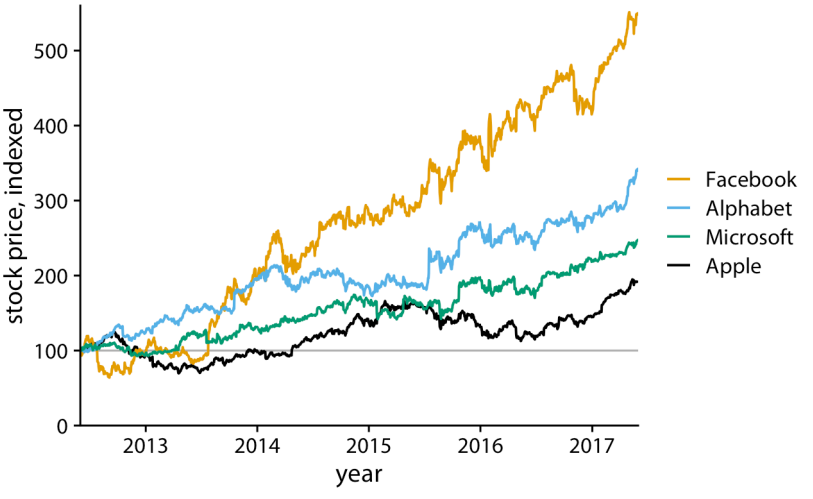


Figure 23-9. Indexed stock price over time for four major tech companies. Adding a thin horizontal line at the index value of 100 to Figure 23-8 helps provide an important reference throughout the entire time period the plot spans. Data source: Yahoo! Finance.

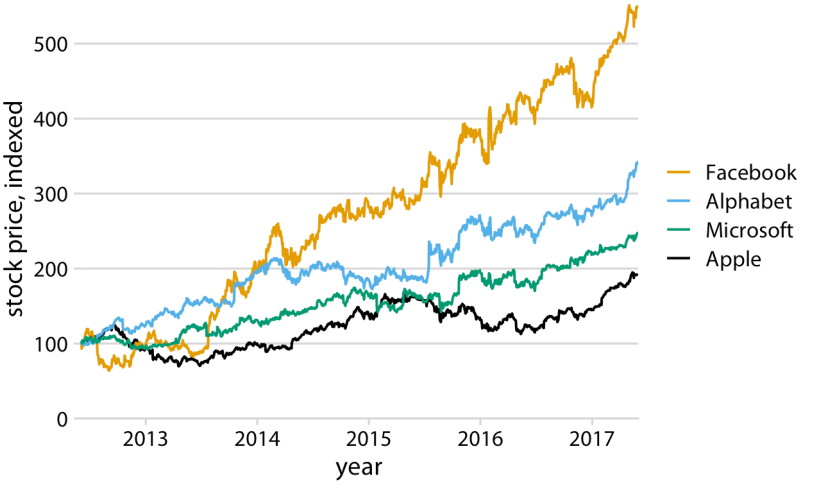


Figure 23-10. Indexed stock price over time for four major tech companies. Adding thin horizontal lines at all major y*-axis ticks provides a better set of reference points than just the one horizontal line of Figure 23-9. This design also removes the need for prominent* x*- and* y*-axis lines, since the evenly spaced horizontal lines create a visual frame for the plot panel. Data source: Yahoo! Finance.*

For such a minimal grid, we generally draw the lines orthogonally to the direction along which the numbers of interest vary. Therefore, if instead of plotting the stock price over time we plot the five-year increase, as horizontal bars, then we will want to use vertical lines instead (Figure 23-11).

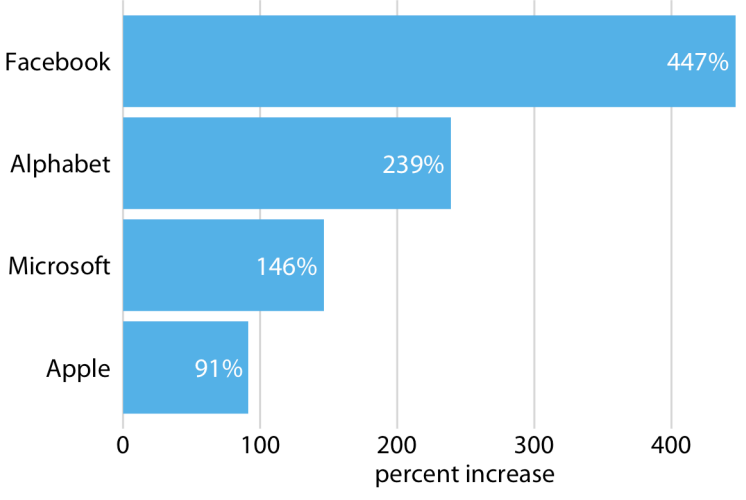


Figure 23-11. Percent increase in stock price from June 2012 to June 2017, for four major tech companies. Because the bars run horizontally, vertical grid lines are appropriate here. Data source: Yahoo! Finance.



Grid lines that run perpendicular to the key variable of interest tend to be the most useful.

For bar graphs such as Figure 23-11, Tufte recommends drawing white grid lines on top of the bars instead of dark grid lines underneath [Tufte 2001]. These white grid lines have the effect of separating the bars into distinct segments of equal length (Figure 23-12). I'm of two minds on this style. On the one hand, research into human perception suggests that breaking bars into discrete segments helps the viewer to perceive bar lengths [Haroz, Kosara, and Franconeri 2015]. On the other hand, to my eye the bars look like they are falling apart and don't form a visual unit. In fact, I used this style purposefully in Figure 6-10 to visually separate stacked bars representing male and female passengers. Which effect dominates may depend on the specific choices of bar width, distance between bars, and thickness of the white grid lines. Thus, if you intend to use this style, I encourage you to vary these parameters until you have a figure that creates the desired visual effect.

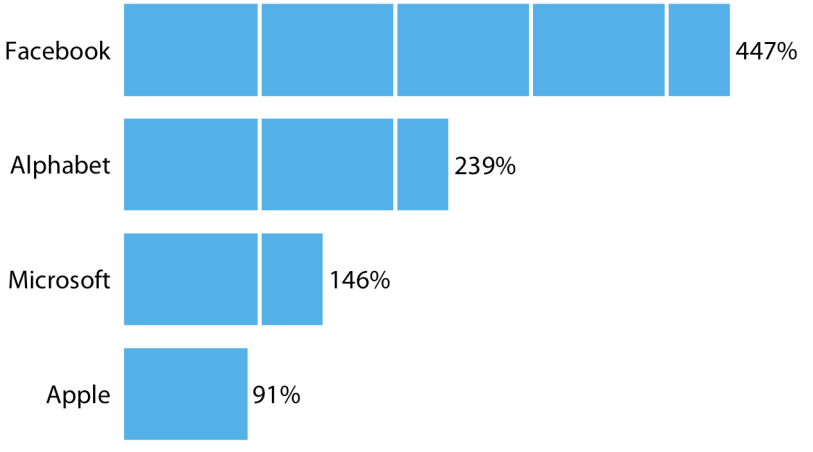


Figure 23-12. Percent increase in stock price from June 2012 to June 2017, for four major tech companies. White grid lines on top of bars can help the reader perceive the relative lengths of the bars. At the same time, they can also create the perception that the bars are falling apart. Data source: Yahoo! Finance.

I would like to point out another downside of Figure 23-12. I had to move the percentage values outside the bars, because the labels didn't fit into the final segments of several of the bars. However, this choice inappropriately visually elongates the bars and should be avoided whenever possible.

Background grids along both axis directions are most appropriate for scatterplots where there is no primary axis of interest. Figure 23-2 at the beginning of this chapter provides an example. When a figure has a full background grid, axis lines are generally not needed.

Paired Data

For figures where the relevant comparison is the $x = y$ line, such as in scatterplots of paired data, I prefer to draw a diagonal line rather than a grid. For example, consider Figure 23-13, which compares gene expression levels in a mutant virus to the nonmutated (wild-type) variant. The diagonal line allows us to see immediately which genes are expressed higher or lower in the mutant relative to the wild type. The same observation is much harder to make when the figure has a background grid and no diagonal line (Figure 23-14). Thus, even though Figure 23-14 looks pleasing, I label it as bad. In particular, gene *10A*, which has a significantly reduced expression level in the mutant relative to the wild-type virus (Figure 23-13), does not visually stand out in Figure 23-14.

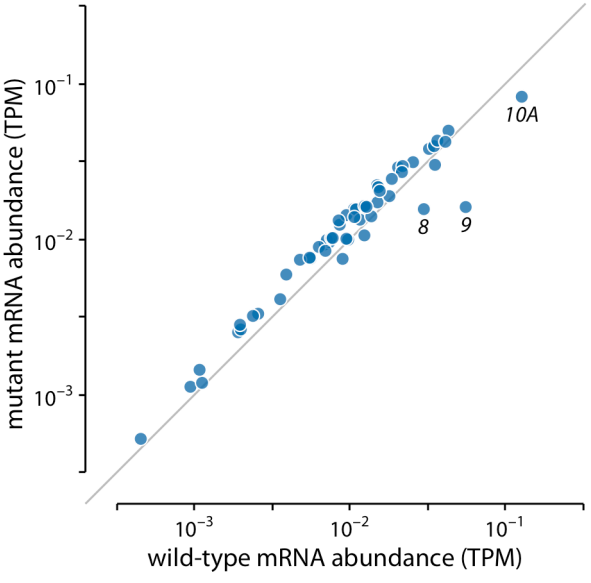


Figure 23-13. Gene expression levels in a mutant bacteriophage T7 relative to wild type. Gene expression levels are measured by mRNA abundances, in transcripts per million (TPM). Each dot corresponds to one gene. In the mutant bacteriophage T7, the promoter in front of gene 9 was deleted, and this resulted in reduced mRNA abundances of gene 9 as well as the neighboring genes 8 and 10A (highlighted). Data source: [Paff et al. 2018].

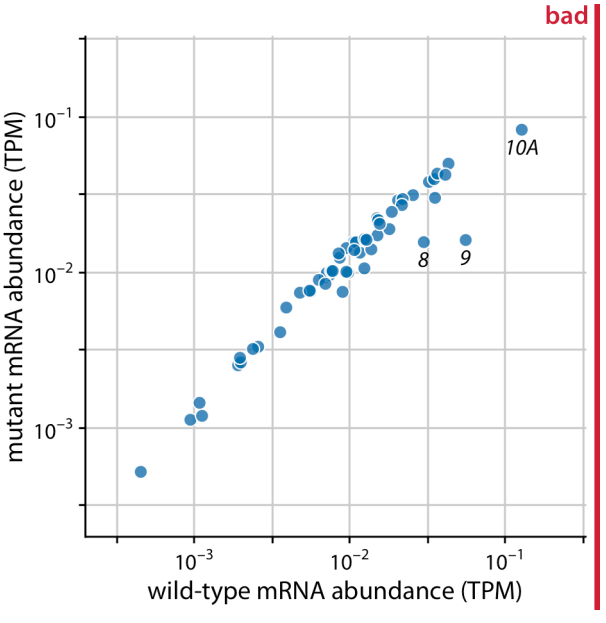


Figure 23-14. Gene expression levels in a mutant bacteriophage T7 relative to wild type. By plotting this dataset against a background grid instead of a diagonal line, we are obscuring which genes are higher or lower in the mutant than in the wild-type bacteriophage. Data source: [Paff et al. 2018].

Of course, we could take the diagonal line from Figure 23-13 and add it on top of the background grid of Figure 23-14, to ensure that the relevant visual reference is present. However, the resulting figure is getting quite busy (Figure 23-15). I had to make the diagonal line darker so it would stand out against the background grid, but now the data points almost seem to fade into the background. We could ameliorate this issue by making the data points larger or darker, but all considered I'd rather choose Figure 23-13.

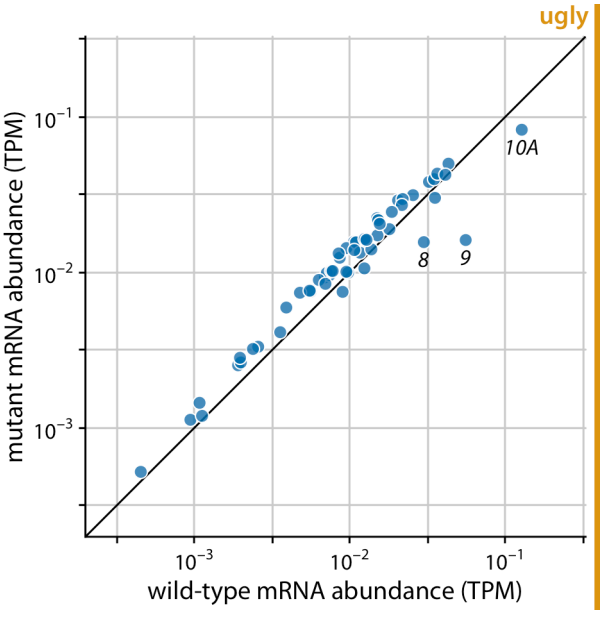


Figure 23-15. Gene expression levels in a mutant bacteriophage T7 relative to wild type. This figure combines the background grid from Figure 23-14 with the diagonal line from Figure 23-13. In my opinion, this figure is visually too busy compared to Figure 23-13, and I would prefer Figure 23-13. Data source: [Paff et al. 2018].

Summary

Both overloading a figure with non-data ink and excessively erasing non-data ink can result in poor figure design. We need to find a healthy medium, where the data points are the main emphasis of the figure while sufficient context is provided about what data is shown, where the points lie relative to each other, and what they mean.

With respect to backgrounds and background grids, there is no one choice that is preferable in all contexts. I recommend being judicious about grid lines. Think carefully about which specific grid or guide lines are most informative for the plot you are making, and then only show those. I prefer minimal, light grids on a white background, since white is the default neutral color on paper and supports nearly any foreground color. However, a shaded background can help the plot appear as a single visual entity, and this may be particularly useful in small multiples plots. Finally, we have to consider how all these choices relate to visual branding and identity. Many magazines and websites like to have an immediately recognizable in-house style, and a shaded background and specific choice of background grid can help create a unique visual identity.

CHAPTER 24

Use Larger Axis Labels

If you take away only one single lesson from this book, make it this one: pay attention to your axis labels, axis tick labels, and other assorted plot annotations. Chances are they are too small. In my experience, nearly all graphing software and plot libraries have poor defaults. If you use the default values, you're almost certainly making a poor choice.

For example, consider Figure 24-1. I see figures like this all the time. The axis labels, axis tick labels, and legend labels are all incredibly small. We can barely see them, and we may have to zoom into the page to read the annotations in the legend.

A somewhat better version of this figure is shown as Figure 24-2. I think the fonts are still too small, and that's why I have labeled the figure as ugly. However, we are moving in the right direction. This figure might be passable under some circumstances. My main criticism here is not so much that the labels aren't legible as that the figure is not balanced; the text elements are too small compared to the rest of the figure.

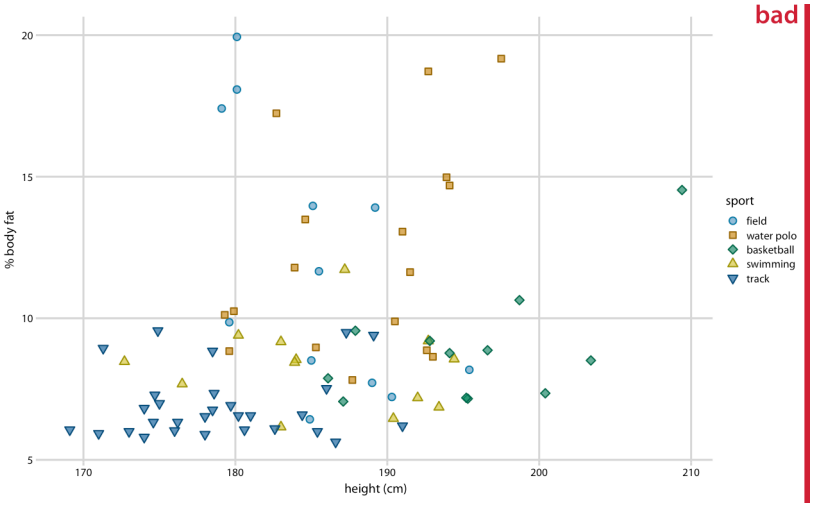


Figure 24-1. Percent body fat versus height in professional male Australian athletes. (Each point represents one athlete.) This figure suffers from the common affliction that the text elements are way too small and are barely legible. Data source: [Telford and Cunningham 1991].

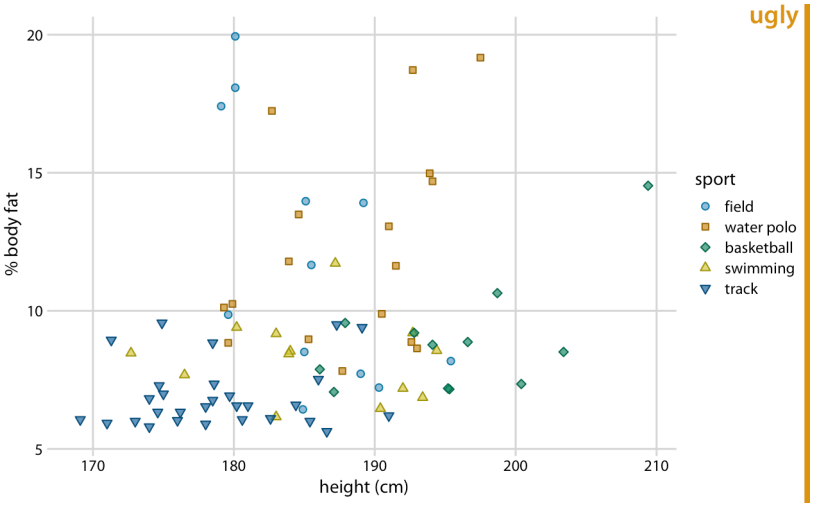


Figure 24-2. Percent body fat versus height in male athletes. This figure is an improvement over Figure 24-1, but the text elements remain too small and the figure is not balanced. Data source: [Telford and Cunningham 1991].

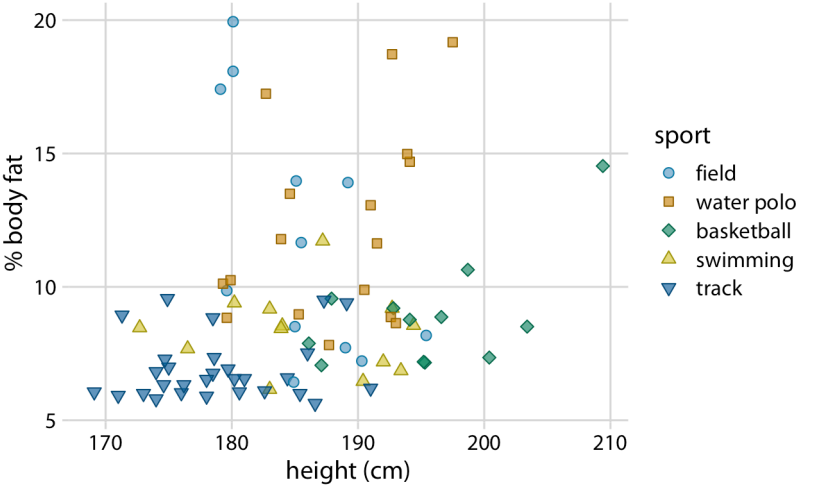


Figure 24-3. Percent body fat versus height in male athletes. All figure elements are appropriately scaled. Data source: [Telford and Cunningham 1991].

Figure 24-3 uses the default settings I'm applying throughout this book. I think it is well balanced; the text is legible, and it fits with the overall size of the figure.

Importantly, we can overdo it and make the labels too big (Figure 24-4). Sometimes we need big labels—for example, if the figure is meant to be reduced in size—but the various elements of the figure (in particular, label text and plot symbols) need to fit together. In Figure 24-4, the points used to visualize the data are too small relative to the text. Once we fix this issue, the figure becomes acceptable again (Figure 24-5).

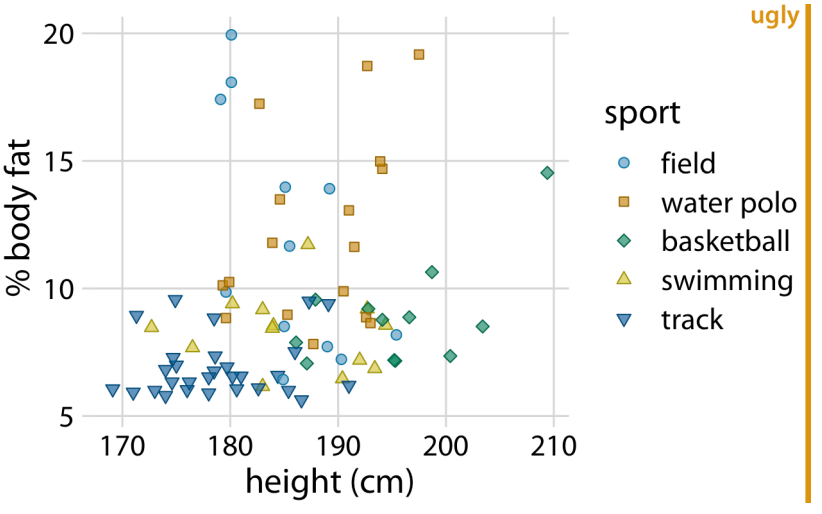


Figure 24-4. Percent body fat versus height in male athletes. The text elements are fairly large, and their size may be appropriate if the figure is meant to be reproduced at a very small scale. However, the figure overall is not balanced; the points are too small relative to the text elements. Data source: [Telford and Cunningham 1991].

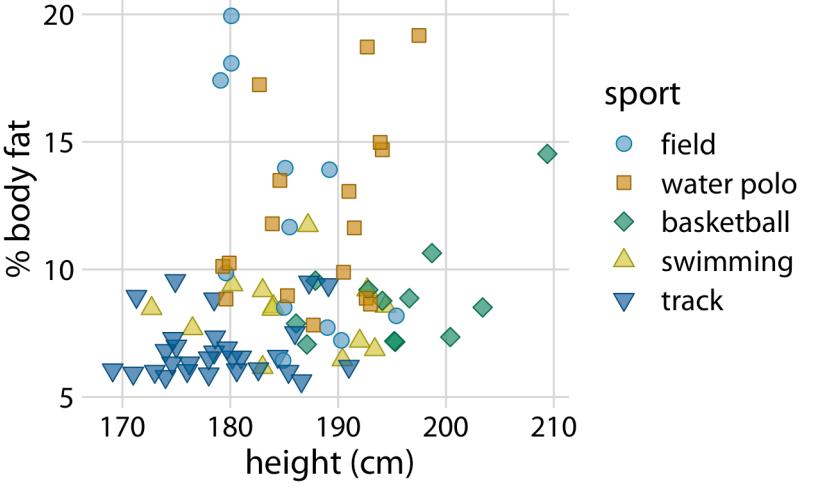


Figure 24-5. Percent body fat versus height in male athletes. All figure elements are sized such that the figure is balanced and can be reproduced at a small scale. Data source: [Telford and Cunningham 1991].

You may look at Figure 24-5 and find everything too big. However, keep in mind that it is meant to be scaled down. Scale the figure down so that it is only two to three inches in width, and it looks just fine. In fact, at that scaling this is the only figure in this chapter that looks good.



Always look at scaled-down versions of your figures to make sure the axis labels are appropriately sized.

I think there is a simple psychological reason for why we routinely make figures whose axis labels are too small, and it relates to large, high-resolution computer monitors. We routinely preview figures on the computer screen, and often we do so while the figure takes up a large amount of space on the screen. In this viewing mode, even comparatively small text seems perfectly fine and legible, and large text can seem awkward and overpowering. In fact, if you take the first figure from this chapter and magnify it to the point where it fills your entire screen, you will likely think that it looks just fine. The solution is to always make sure that you look at your figures at a realistic print size. You can either zoom out so they are only three to five inches in width on your screen, or stand well back and check whether the figure still looks good from a substantial distance.

CHAPTER 25
Avoid Line Drawings

Whenever possible, visualize your data with solid, colored shapes rather than with lines that outline those shapes. Solid shapes are more easily perceived as coherent objects, are less likely to create visual artifacts or optical illusions, and more immediately convey amounts than do outlines. In my experience, visualizations using solid shapes are both clearer and more pleasant to look at than equivalent versions that use line drawings. Thus, I avoid line drawings as much as possible. However, I want to emphasize that this recommendation does not supersede the principle of proportional ink (Chapter 17).

Line drawings have a long history in the field of data visualization because throughout most of the 20th century, scientific visualizations were drawn by hand and had to be reproducible in black and white. This precluded the use of areas filled with solid colors, including solid grayscale fills. Instead, filled areas were sometimes simulated by applying hatch, cross-hatch, or stipple patterns. Early plotting software imitated the hand-drawn simulations and similarly made extensive use of line drawings, dashed or dotted line patterns, and hatching. While modern visualization tools and modern reproduction and publishing platforms have none of the earlier limitations, many plotting applications still default to outlines and empty shapes rather than filled areas. To raise your awareness of this issue, here I'll show you several examples of the same figures drawn with both lines and filled shapes.

The most common and at the same time most inappropriate use of line drawings is seen in histograms and bar plots. The problem with bars drawn as outlines is that it is not immediately apparent which side of any given line is inside a bar and which side is outside. As a consequence, in particular when there are gaps between bars, we end up with a confusing visual pattern that detracts from the main message of the figure (Figure 25-1). Filling the bars with a light color, or with gray if color reproduction is not possible, avoids this problem (Figure 25-2).

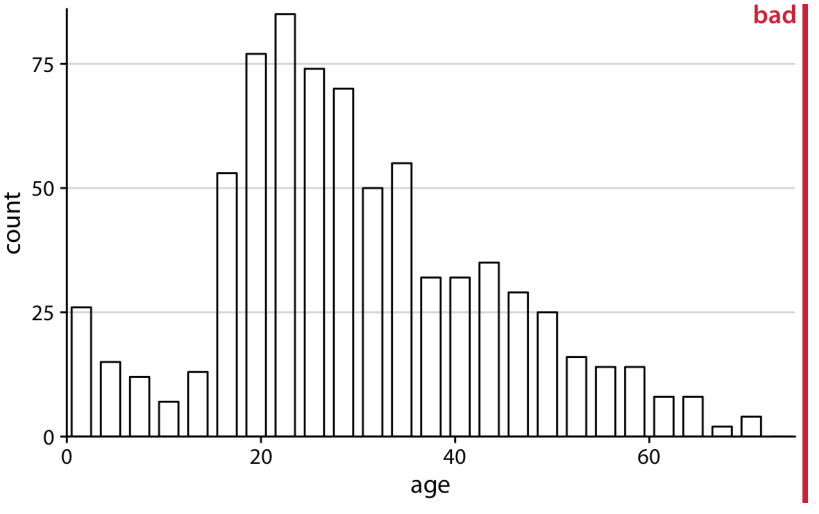


Figure 25-1. Histogram of the ages of Titanic *passengers, drawn with empty bars. The empty bars create a confusing visual pattern. In the center of the histogram, it is difficult to tell which parts are inside of bars and which parts are outside. Data source: Encyclopedia Titanica.*

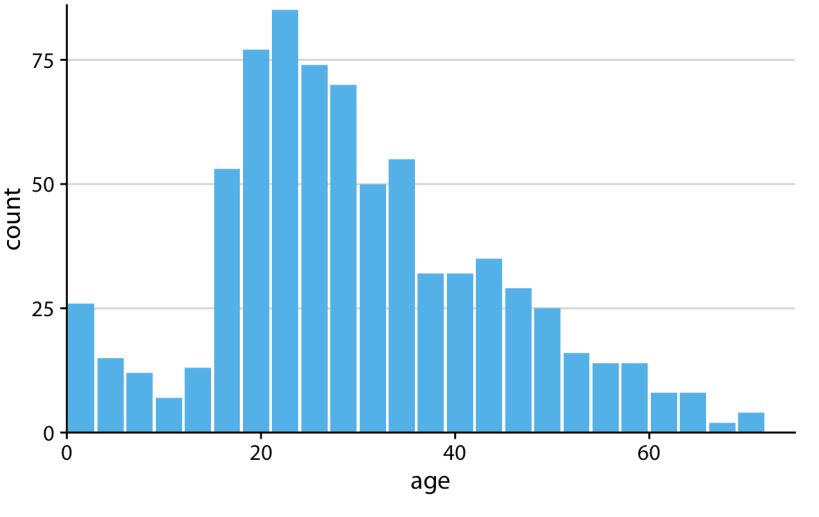


Figure 25-2. Histogram of the ages of Titanic *passengers. This is the same histogram as in Figure 25-1, now drawn with filled bars. The shape of the age distribution is much more easily discernible in this variation of the figure. Data source: Encyclopedia Titanica.*

Next, let's take a look at an old-school density plot. I'm showing density estimates for the sepal length distributions of three species of *Iris*, drawn entirely in black and white as a line drawing (Figure 25-3). The distributions are shown just by their outlines, and because the figure is in black and white I'm using different line styles to distinguish them. This figure has two main problems. First, the dashed line styles do not provide a clear separation between the area under the curve and the area above it. While the human visual system is quite good at connecting the individual line elements into a continuous line, the dashed lines nevertheless look porous and do not serve as a strong boundary for the enclosed area. Second, because the lines intersect and the areas they enclose are not shaded, it is difficult to segment the different densities from the six distinct shape outlines. This effect would have been even stronger had I used solid rather than dashed lines for all three distributions.

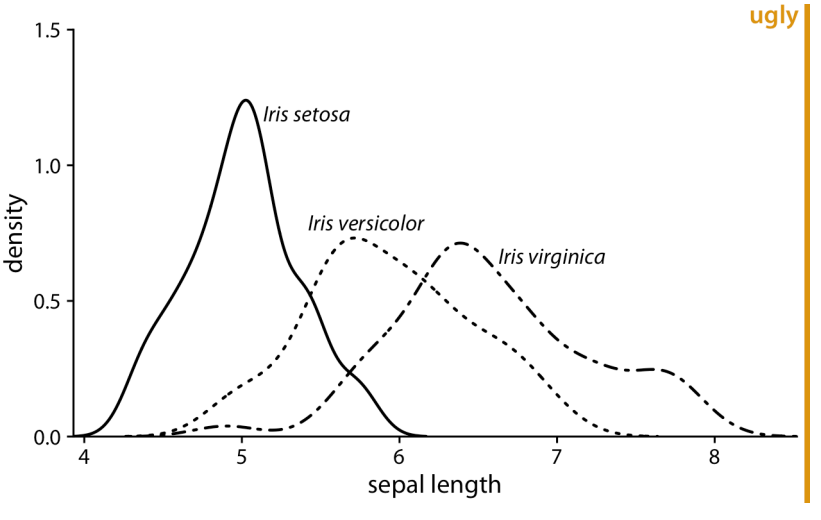


Figure 25-3. Density estimates of the sepal lengths of three different Iris *species. The broken line styles used for* Iris versicolor *and* Iris virginica *detract from the perception that the areas under the curves are distinct from the areas above them. Data source: [Fisher 1936].*

We can attempt to address the problem of porous boundaries by using colored lines rather than dashed lines (Figure 25-4). However, the density areas in the resulting plot still have little visual presence. Overall, I find the version with filled areas (Figure 25-5) the most clear and intuitive. It is important, however, to make the filled areas partially transparent, so that the complete distribution for each species is visible.

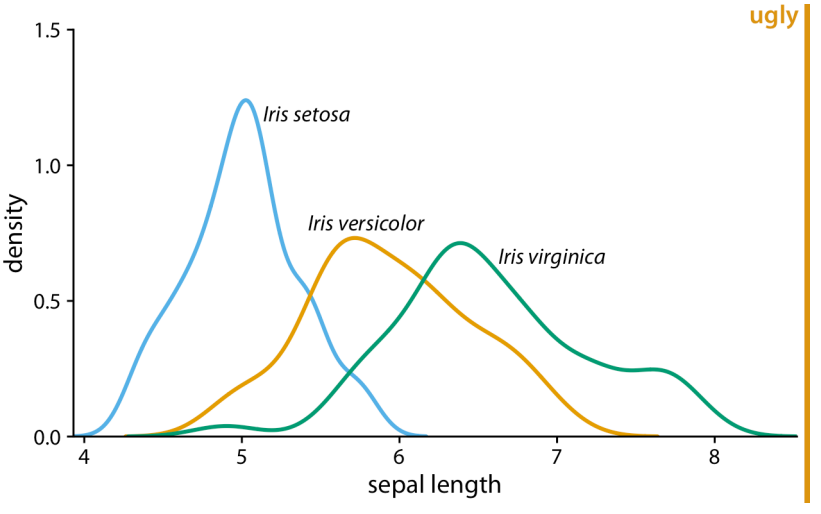


Figure 25-4. Density estimates of the sepal lengths of three different Iris *species. By using solid, colored lines we have solved the problem of Figure 25-3 that the areas below and above the lines seem to be connected. However, we still don't have a strong sense of the size of the area under each curve. Data source: [Fisher 1936].*

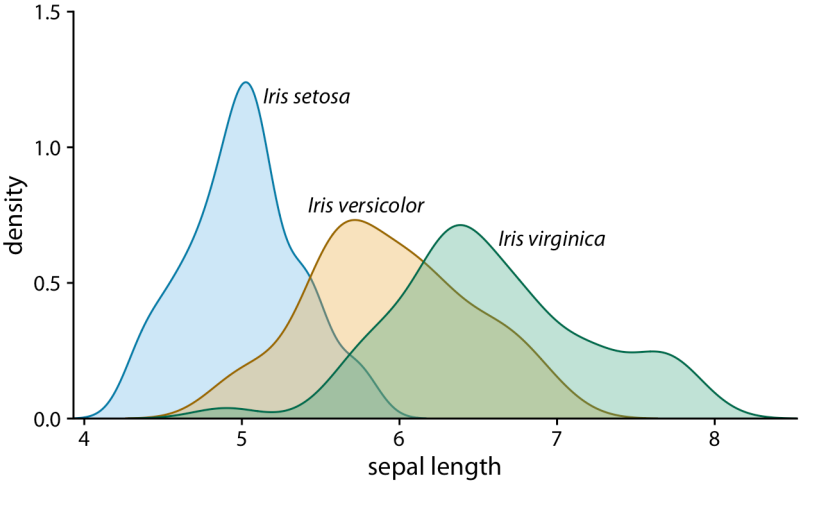


Figure 25-5. Density estimates of the sepal lengths of three different Iris *species, shown as partially transparent shaded areas. The shading helps us perceive the three density curves as three distinct objects. Data source: [Fisher 1936].*

Line drawings also arise in the context of scatterplots, when different point types are drawn as open circles, triangles, crosses, etc. As an example, consider Figure 25-6. The figure contains a lot of visual noise, and the different point types do not strongly separate from each other. Drawing the same figure with solidly colored shapes addresses this issue (Figure 25-7).

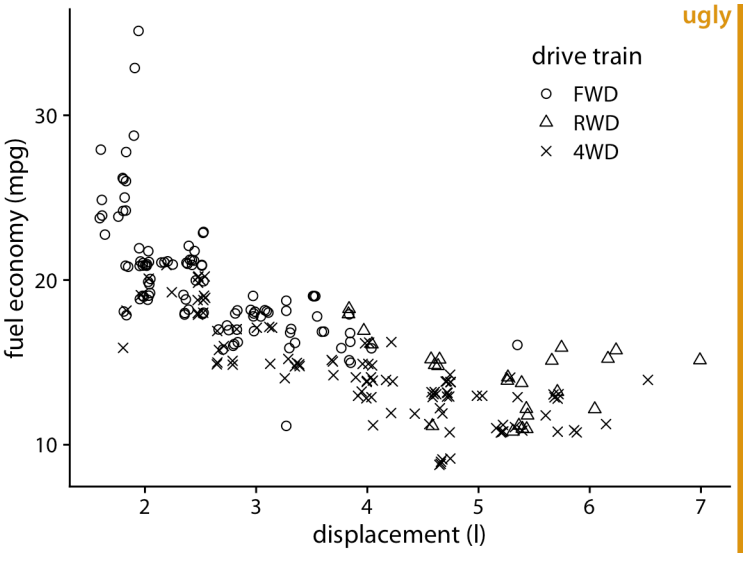


Figure 25-6. City fuel economy versus engine displacement, for cars with front-wheel drive (FWD), rear-wheel drive (RWD), and all-wheel drive (4WD). The different point styles, all black-and-white line-drawn symbols, create substantial visual noise and make it difficult to read the figure. Data source: US Environmental Protection Agency (EPA), https://fueleconomy.gov.

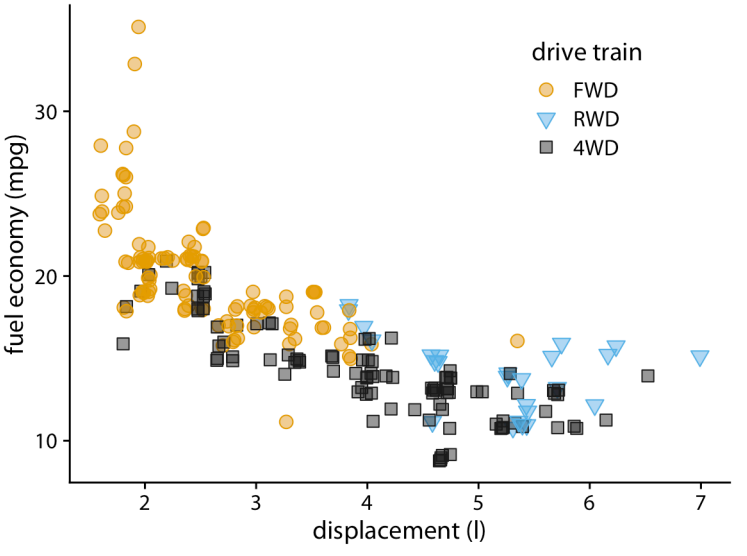


Figure 25-7. City fuel economy versus engine displacement. By using both different colors and different solid shapes for the different drive-train variants, this figure visually separates the drive-train variants while remaining reproducible in grayscale if needed. Data source: EPA.

I strongly prefer solid points over open points, because the solid points have much more visual presence. The argument that I sometimes hear in favor of open points is that they help with overplotting, since the empty areas in the middle of each point allow us to see other points that may be lying underneath. In my opinion, the benefit of being able to see overplotted points does not, in general, outweigh the detriment of the added visual noise of open symbols. There are other approaches for dealing with overplotting; see Chapter 18 for some suggestions.

Finally, let's consider boxplots. Boxplots are commonly drawn with empty boxes, as in Figure 25-8. I prefer a light shading for the box, as in Figure 25-9. The shading separates the box from the plot background, and it helps in particular when we're showing many boxplots right next to each other, as is the case in Figures 25-8 and 25-9. In Figure 25-8, the large number of boxes and lines can again create the illusion of background areas outside of boxes being actually on the inside of some other shape, just as we saw in Figure 25-1. This problem is eliminated in Figure 25-9. I have sometimes heard the critique that shading the inside of the box gives too much weight to the center 50% of the data, but I don't buy that argument. It is inherent to the boxplot, shaded box or not, to give more weight to the center 50% of the data than to the rest. If you don't want this emphasis, then don't use a boxplot. Instead, use a violin plot, jittered points, or a sina plot (Chapter 9).

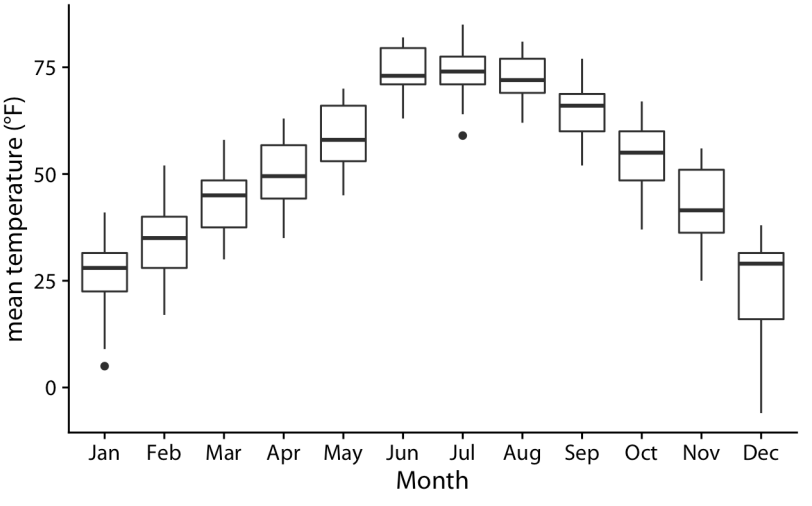


Figure 25-8. Distributions of daily mean temperatures in Lincoln, NE, in 2016. Boxes are drawn in the traditional way, without shading. Data source: Weather Underground.

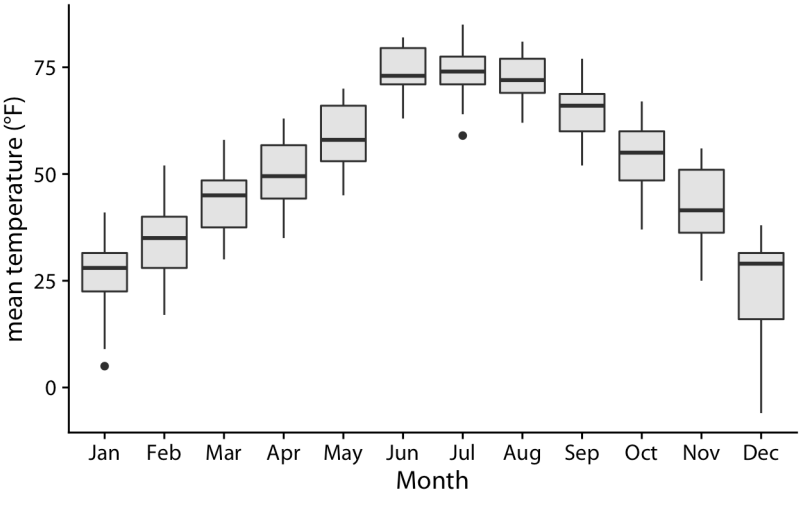


Figure 25-9. Distributions of daily mean temperatures in Lincoln, NE, in 2016. By giving the boxes a light gray shading, we can make them stand out better against the background. Data source: Weather Underground.